

Where we are

Assignment 1: graded ~65/84, expecting Tuesday or Wednesday

Assignment 2: due this Thursday

Lab 6 is in-class activity

Assignment 3 will be released this Friday (due 2 weeks after spring break)

Extra credit assignment: 1 hour for 1% or \$25; by March 14 but can extend

Storytelling & Narrative

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

March 10, 2025

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

Using data to craft a
compelling **narrative**,
often involving **interactivity**

Value of storytelling

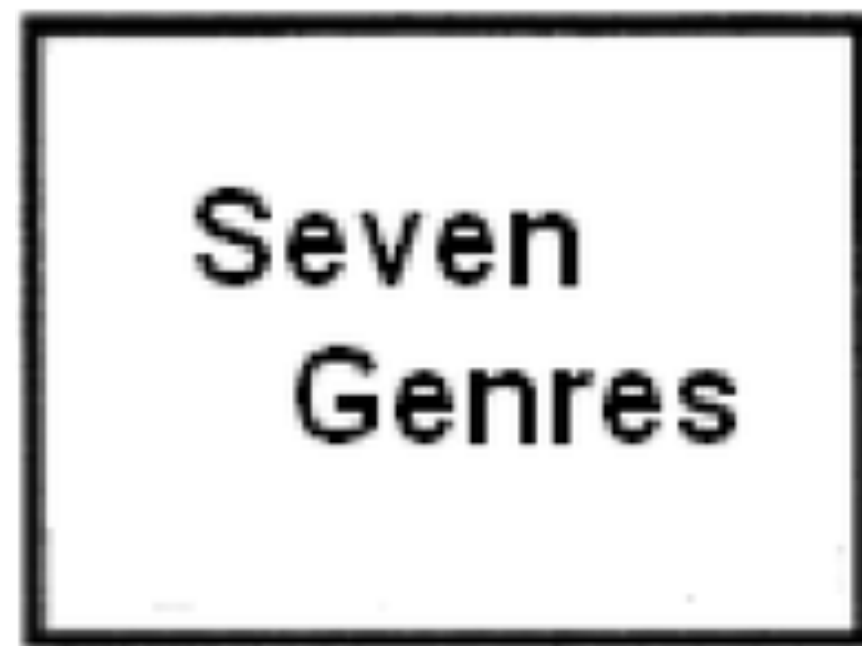
Get beyond **exploring** data to **engaging** and **explaining**



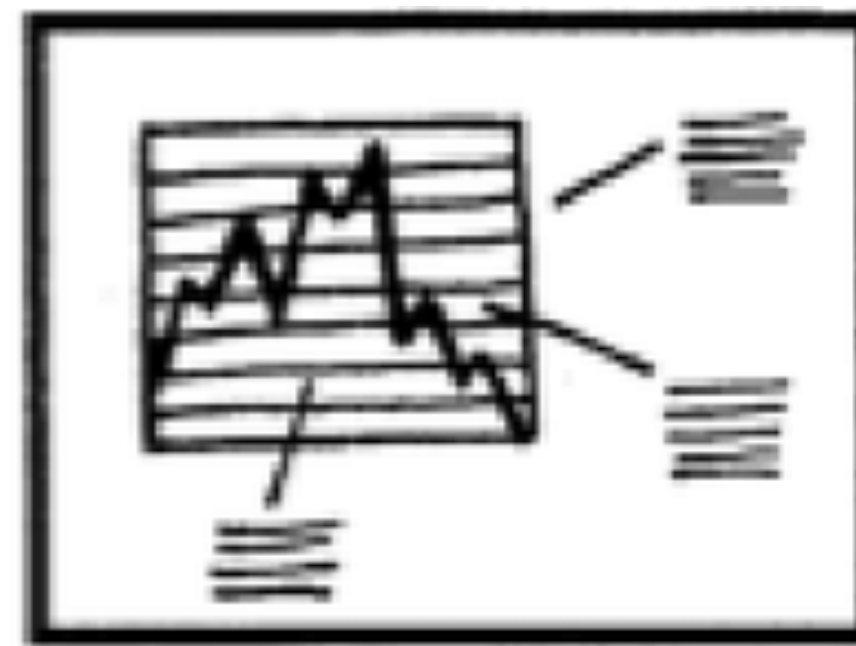
interactivity in storytelling
designer/reader balance

Engages through **conflict, resolution, uncertainty**

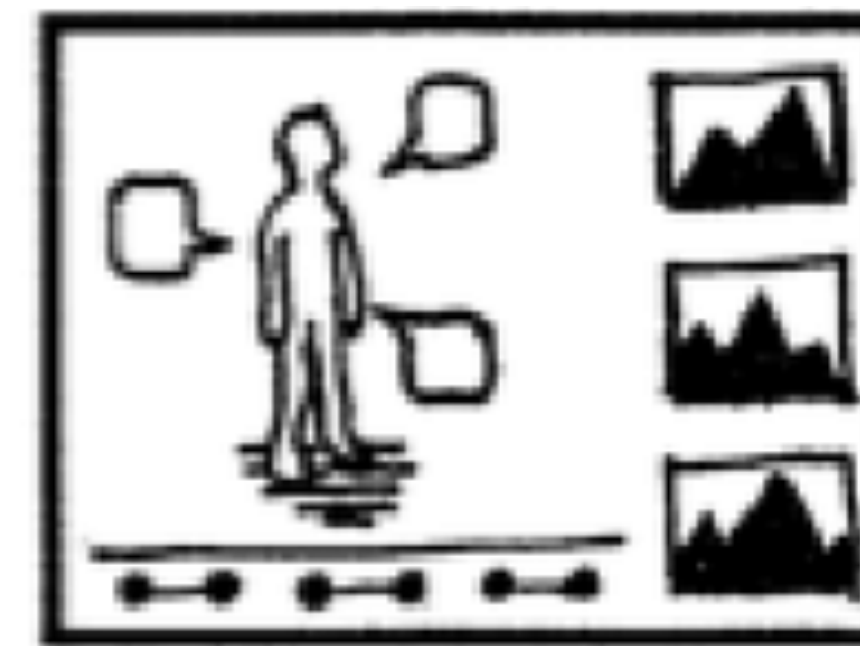
Narrative genres



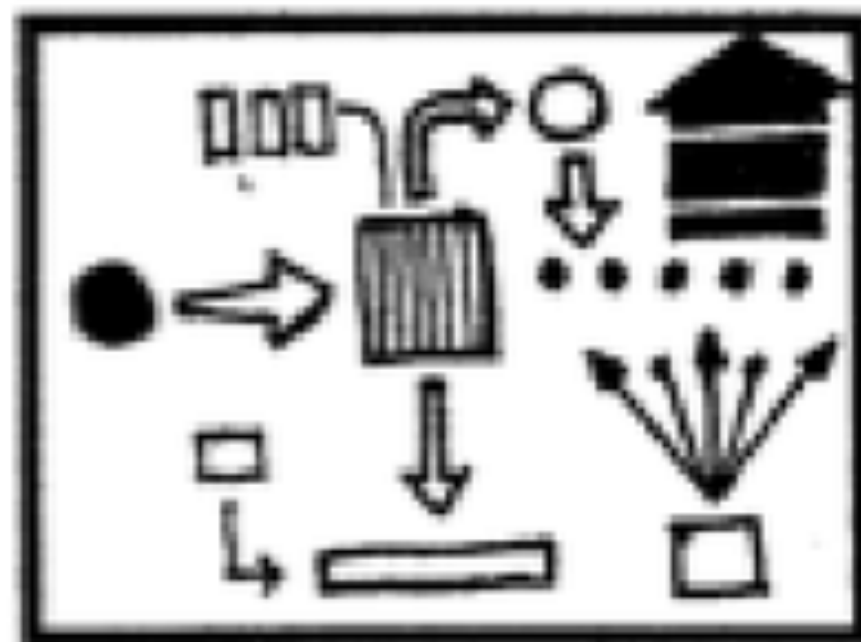
Magazine Style



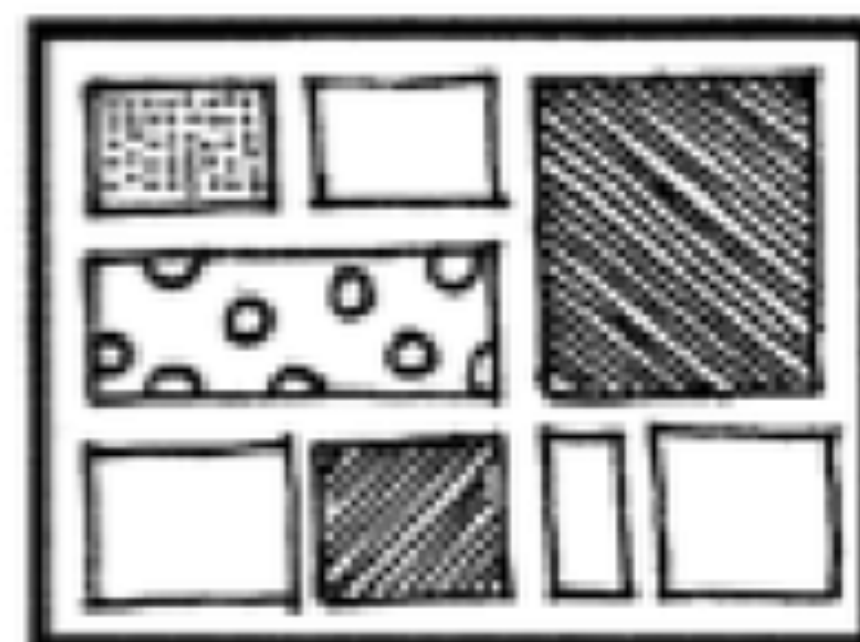
Annotated Chart



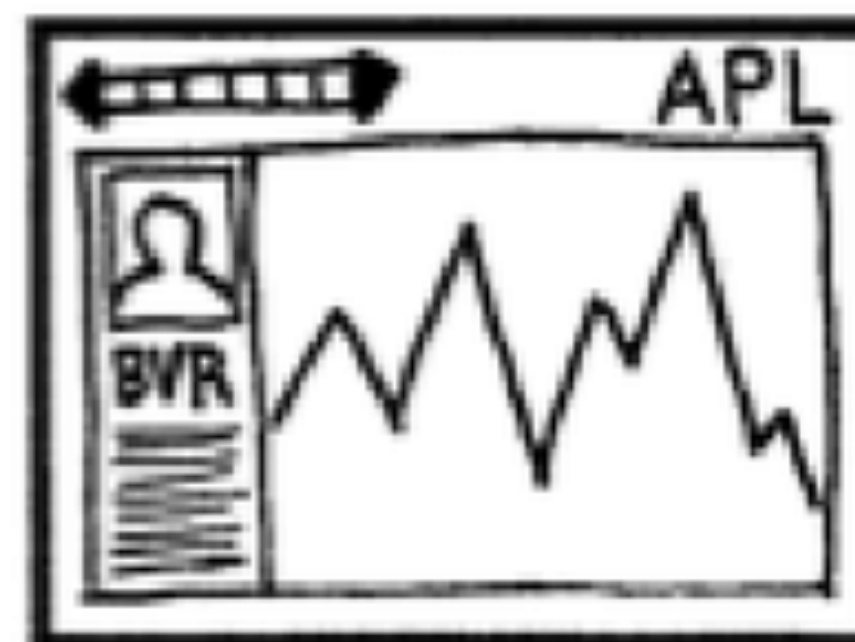
Partitioned Poster



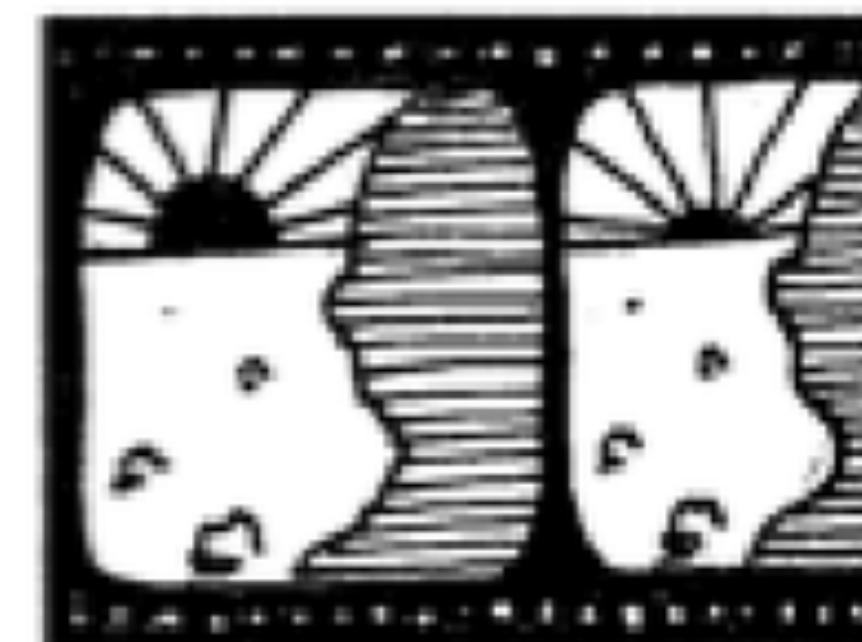
Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

Not mutually exclusive; differ in **ordering** and **designer/reader control**

Magazine style

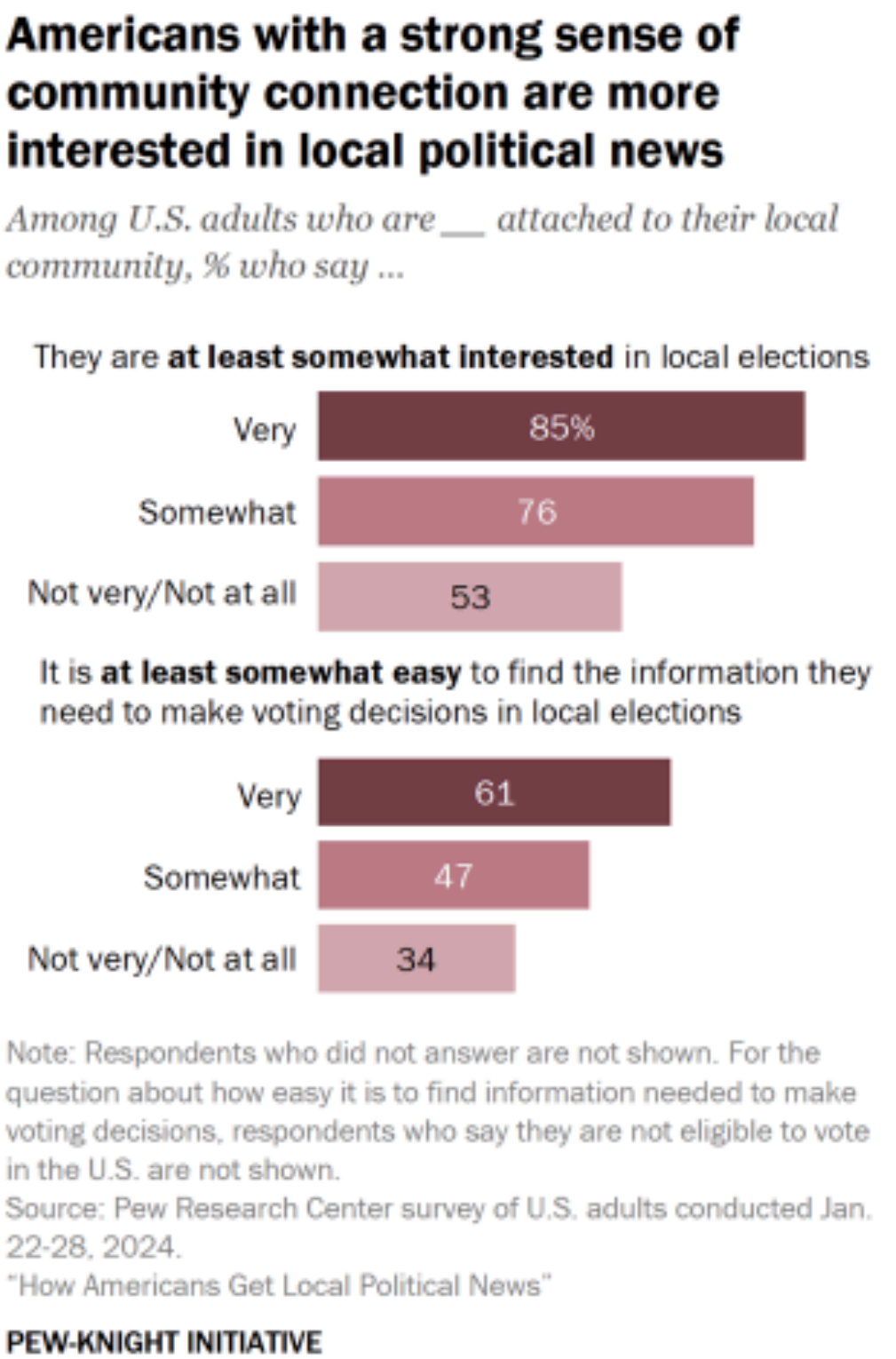
An image embedded
in a page of text



Americans who are more attached to their local community are more interested in and satisfied with local political news

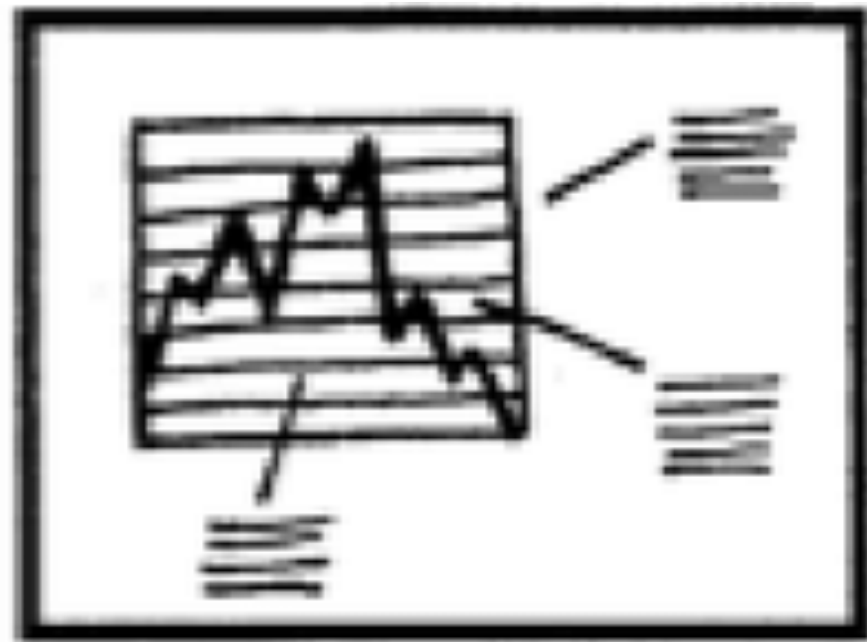
There are clear links between Americans’ sense of community attachment and their experiences with news on local government and politics. Those who feel more attached to their community have more interest in news about local laws and local elections, higher satisfaction with the quality of their local political news, and an easier time finding the information they need to vote.

- 91% of Americans who feel very attached to their community are at least somewhat interested in news about local laws and policies, compared with 68% of those who are not very or not at all attached. A similar pattern holds for interest in news about local elections.
- A majority of Americans with a strong sense of community connection (61%) say it is at least somewhat easy to find the information they need to make voting decisions in local elections. Among those with little or no sense of community attachment, just 34% feel this way.
- Among U.S. adults who follow news about local government and politics, four-in-ten of those with strong community attachment are extremely or very satisfied with the local political news they get – more than double the share among those who feel little or no attachment to their community (16%).



Annotated chart

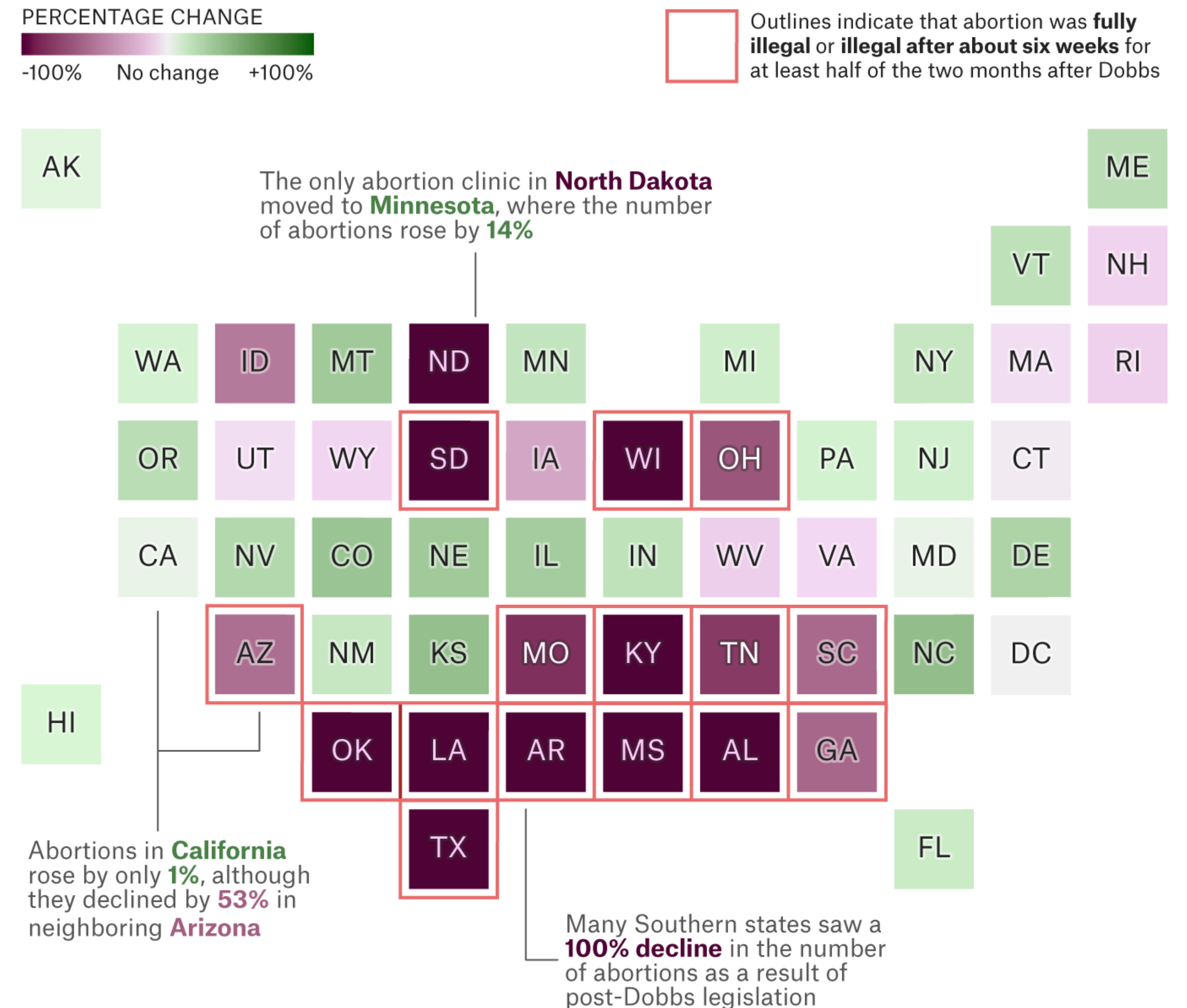
Usually with a catchy title
to tell the key point



The Dobbs v. Jackson Women's Health Organization ruling was a 2022 Supreme Court decision that overturned the constitutional right to abortion and eliminated federal standards on abortion access.

Several states saw a complete drop in abortions after Dobbs

Percentage change in number of abortions from April to August 2022, by state



Percent change in abortions is calculated from #WeCount's unrounded data. Less-populated states might see larger percentage shifts because the raw number of abortions in those states tend to be smaller.

Usually with a catchy title
to tell the key point



Iraq's bloody toll

Monthly fatalities 0 500 1,000 1,500 2,000 2,500 3,000 3,500 3,977

2003 2004 2005 2006 2007 2008 2009 2010 2011

Coalition deaths
4,802

Civilian deaths
113,728

December
Saddam Hussein captured

May
US President George W. Bush declares an end to combat

Jun 2004
US hands sovereignty to interim Iraqi government

December, 2005
Iraqis elect full-term government and parliament

August, 2005
950 killed in a single incident when fears of an insurgent attack caused a huge crowd of Shiite pilgrims to stampede on a bridge

January, 2007
Bush announces a surge of extra troops will be deployed

December, 2006
Saddam Hussein evicted

January, 2009
US forces come under Iraqi mandate. Control of Green Zone handed to Iraq

March, 2003
US-led forces invade

The biggest killers
How civilians died, according to the data available*

Invasion to March 29, 2008

Air strikes 3,050

Roadside bombs 2,854

Salicide bombs 8,708

Vehicle bombs 5,360

Mortar fire 2,809

Executions 19,306

Small arms gunfire 11,877

US ground troops in Iraq

200 thousand

150 thousand

100 thousand

50 thousand

0

2003 2004 2005 2006 2007 2008 2009 2010 2011

President Bush approves 21,500 extra troops

Coalition deaths

United States 4,484

Others 129

Britain 179

Total
4,802

Iraqi security forces
10,125
to July 2010

Coalition fatalities by area
Baghdad and the east desert province of Anbar saw some of the fiercest fighting, the latter being the scene of sectarian tensions and a bloody Sunni insurgency

0-99

100-499

500+

Dahuk 0

Ninawa 0

Arbil 2

Kirkuk 106

Salahaddin 437

Diwala 266

Baghdad 1,421

Anbar 1,335

Karbala 38

Babil 217

Wasit 52

Quinniyah 43

Najaf 33

Obi Qar 99

Muthana 8

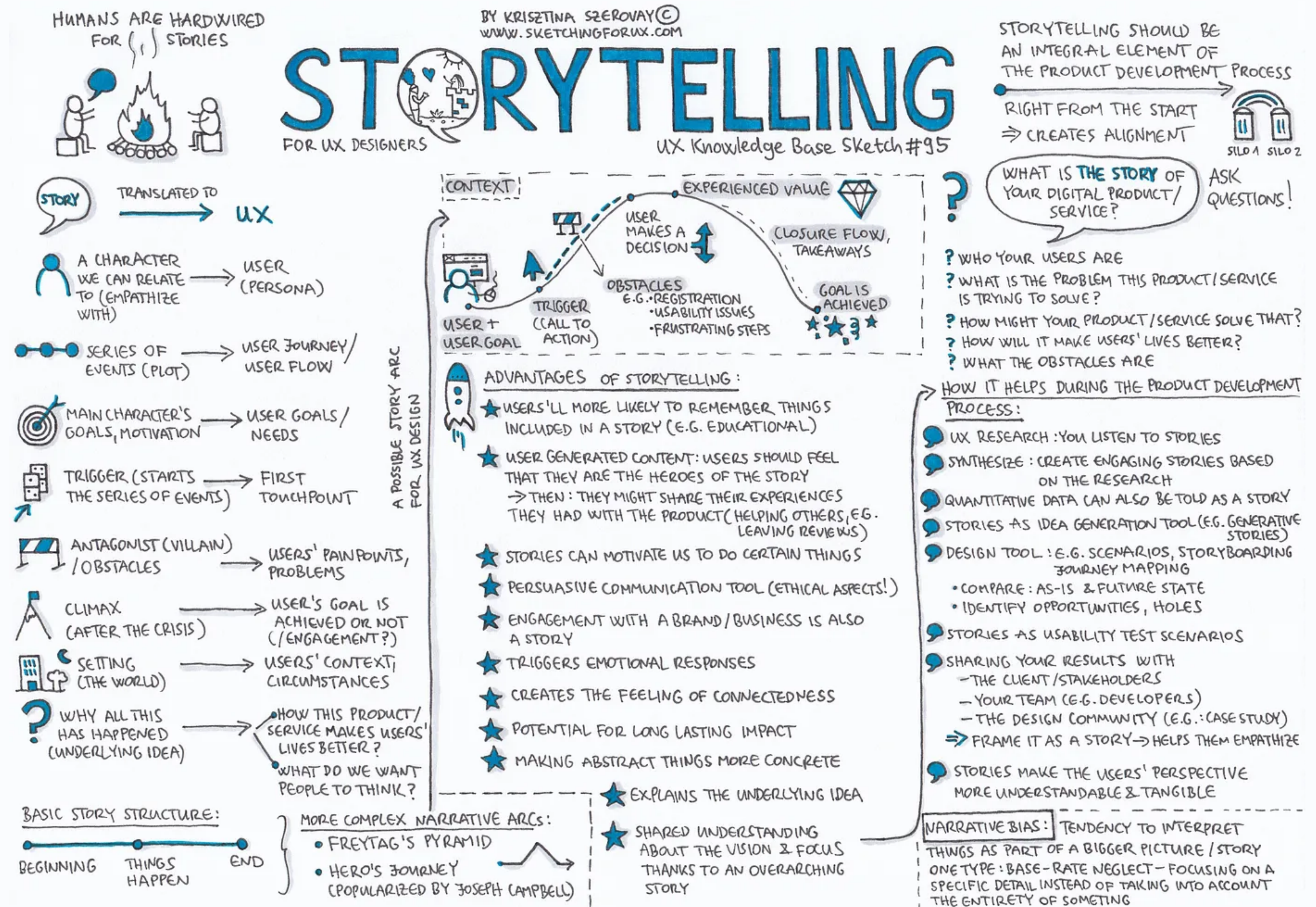
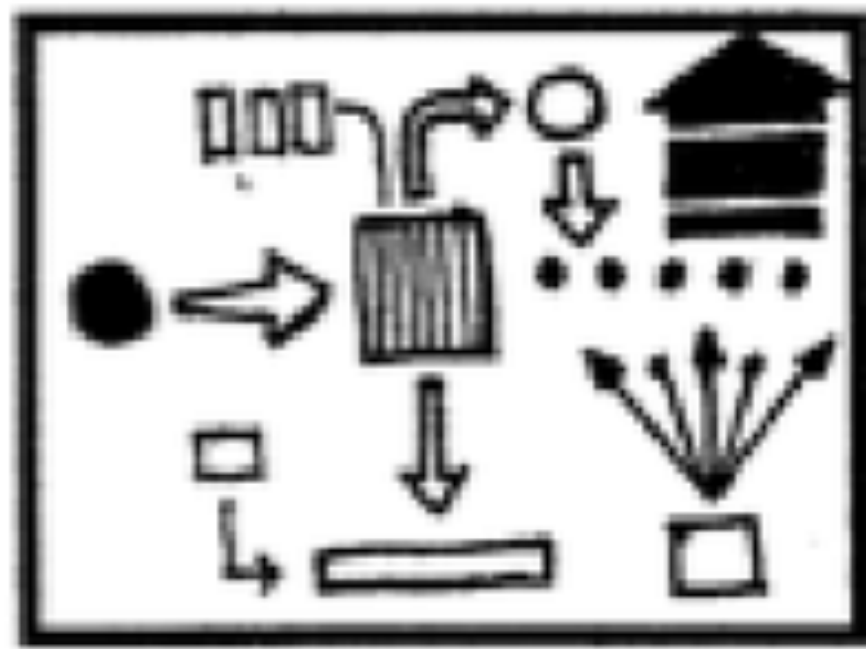
Maysan 29

Basra 156

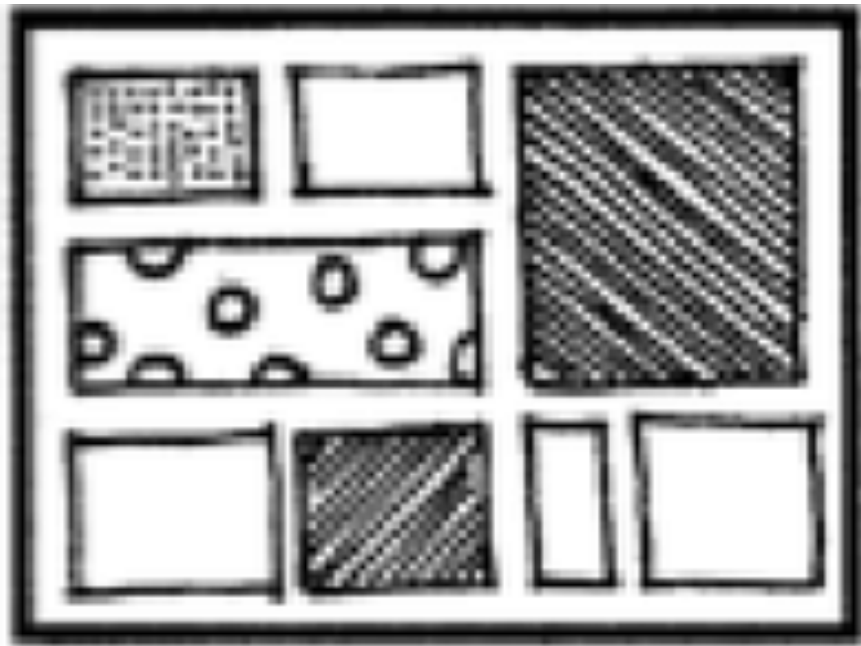
*Source: Reuters, Associated Press, and other sources. Figures are approximate and may change as more data becomes available.

Flow chart

For lack of a better example....



Comic strip



AN ILLUSTRATED GUIDE TO DATA LITERACY

PIE CHARTS

GOOD MORNING!

HEY BECKY!

BEFORE THE MEETING, I NEED TO SEE WHICH STATE IS THE BIGGEST PROPORTION OF OUR SALES.

SURE...

psst, BECKY... PROPORTION JUST MEANS A PART COMPARED TO THE WHOLE.

SURE...

MAYBE A PIE CHART? BUT AREN'T THEY BAD?

THEY SHOW PROPORTIONS WELL... REMEMBER ALL SLICES MUST ADD UP TO 100%.

OH NO, THAT SEEMS LIKE TOO MANY SLICES!

I WISH WE HAD TRIED A DIFFERENT CHART...

HOW'D IT GO?

SHE COULDN'T TELL WHICH SLICE WAS BIGGEST... SMALL DIFFERENCES ARE LOST IN A PIE CHART...

IT'D WORK BETTER FOR PROPORTIONS THAT ARE MORE RECOGNIZABLE: 25% 50% 75%

WAFFLES IN THE BREAK ROOM! THEY RAISED THE BAR TODAY!

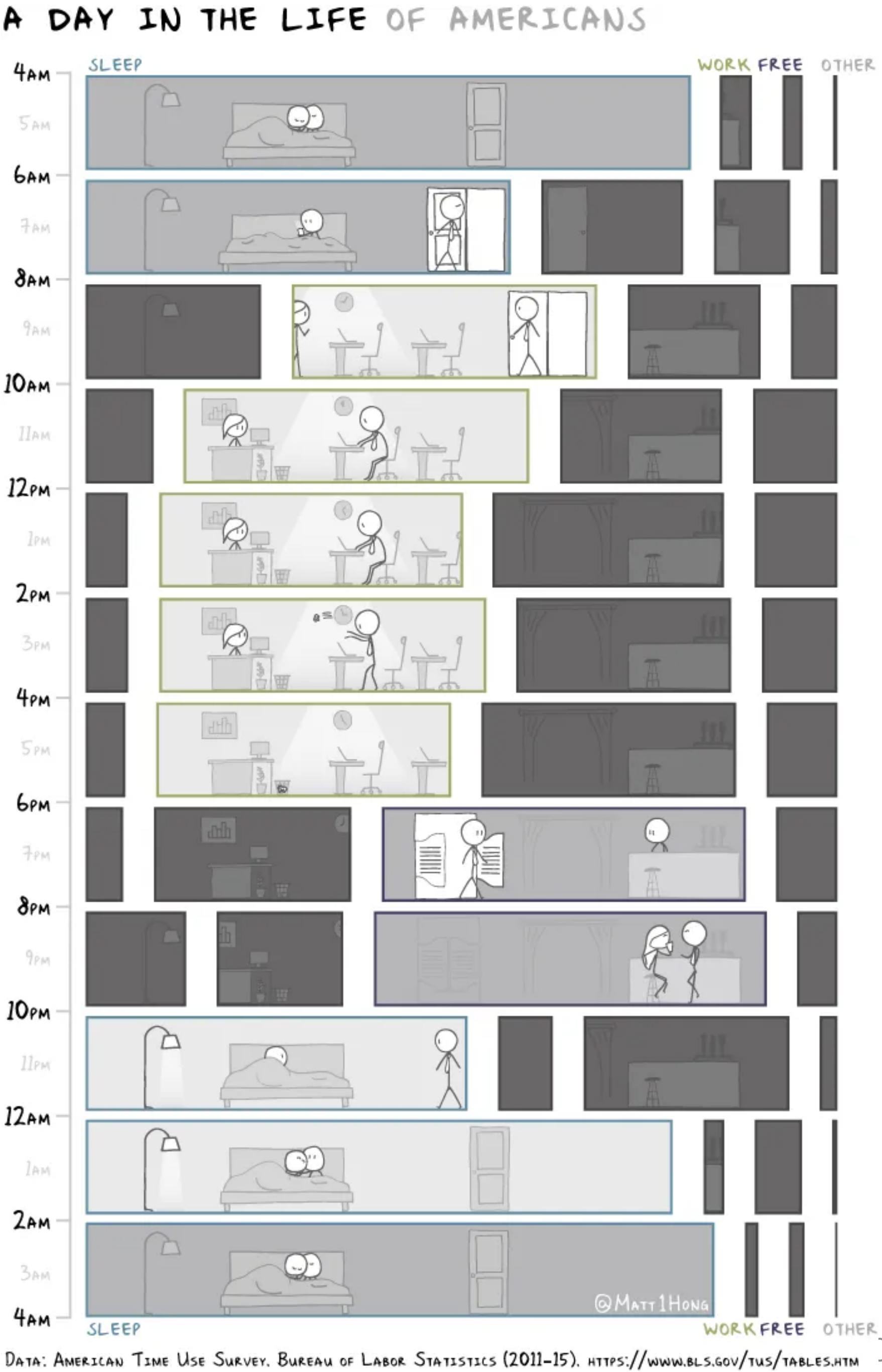
PROPORTION OF SALES BY STATE

20% 18% 15% 15% 13% 9% 5% 5%

VA GA MD NC SC FL NY NJ

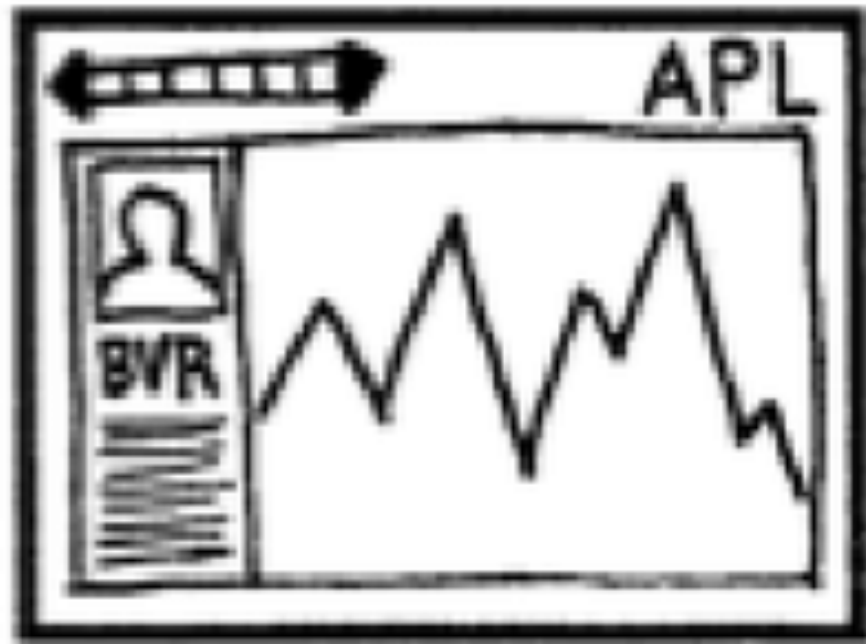
WE SEE THAT VIRGINIA IS THE BIGGEST PROPORTION OF SALES.

THE PIE CHART: A GOOD CHOICE WHEN YOU'RE SHOWING A PART-TO-WHOLE RELATIONSHIP WITH ONLY A FEW SLICES THAT HAVE LARGE DIFFERENCES. ALL SLICES SHOULD ADD UP TO 100%, OTHERWISE YOU MAY HAVE ADDED OR REMOVED A CATEGORY, OR IT'S NOT REALLY A PART-TO-WHOLE RELATIONSHIP. OTHER CHART TYPES TO TRY: BAR OR STACKED BAR CHART, TREEMAP, OR A WAFFLE CHART. NOT ALL PIE CHARTS ARE BAD; JUST BE THOUGHTFUL WITH YOUR CHOICES.



Slide show

Often with
interactivity



g

Scientific Proof that Americans are Completely Addicted to Trucks

By Adam Pearce Blacki Migliozi and David Ingold | Jan. 15, 2015

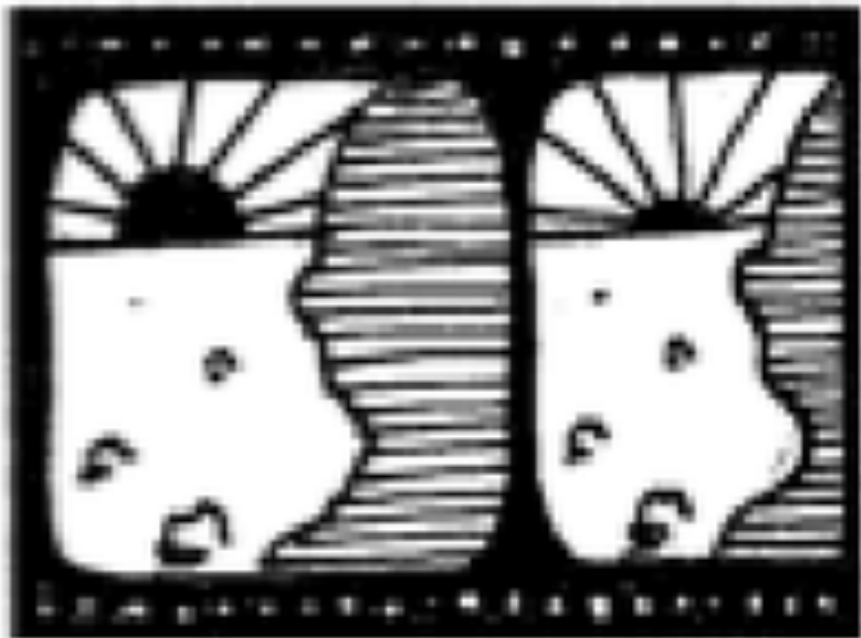
Car geeks will be elbowing each other out of the way for a glimpse of Ford's gorgeous new GT sports car when the Detroit auto show opens to the public this weekend. But when it comes to the new vehicles people actually buy, the story is all about light-trucks—the broad category that includes pickups, SUVs, crossovers, and minivans. As the economy gains strength and gas prices drop, Americans are turning away from cars and snapping up trucks of all sizes.

2014 new U.S. auto sales, by model

Scroll



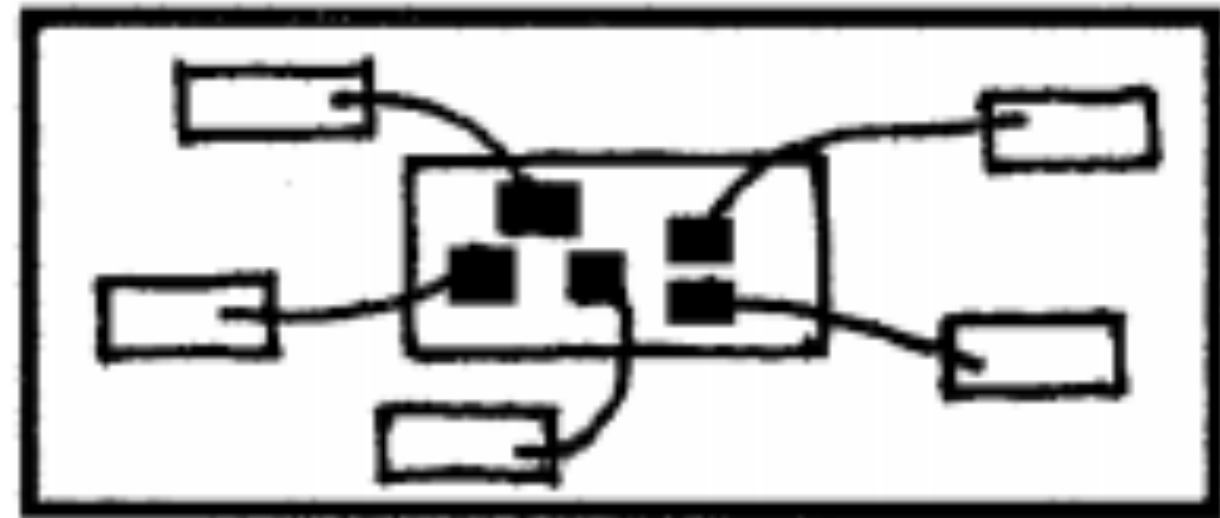
Film/Video



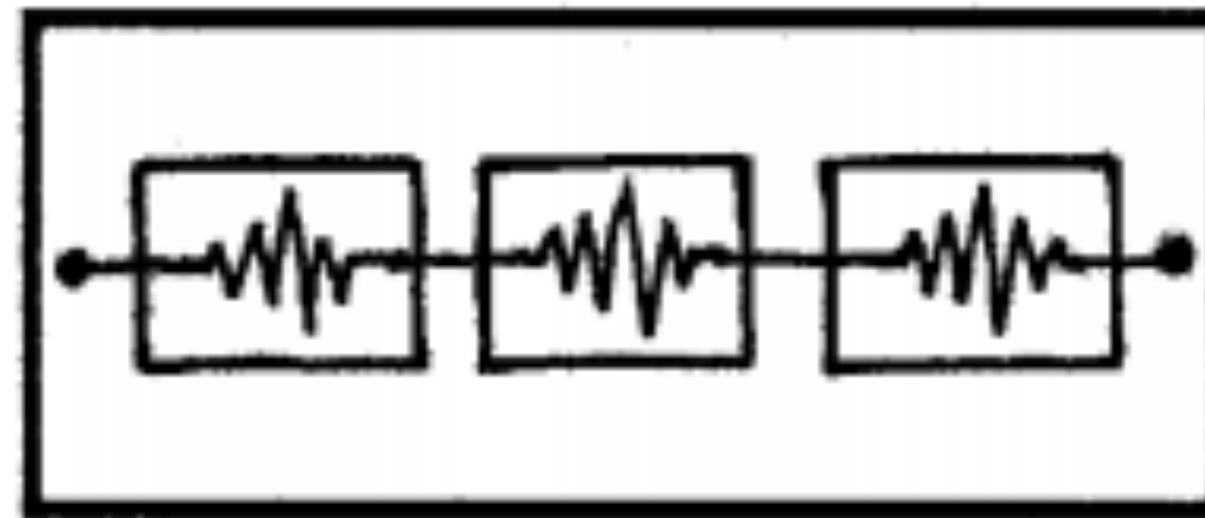
Slideshow might be the most
common and engaging format...

Techniques for storytelling: 3 types of interactivity

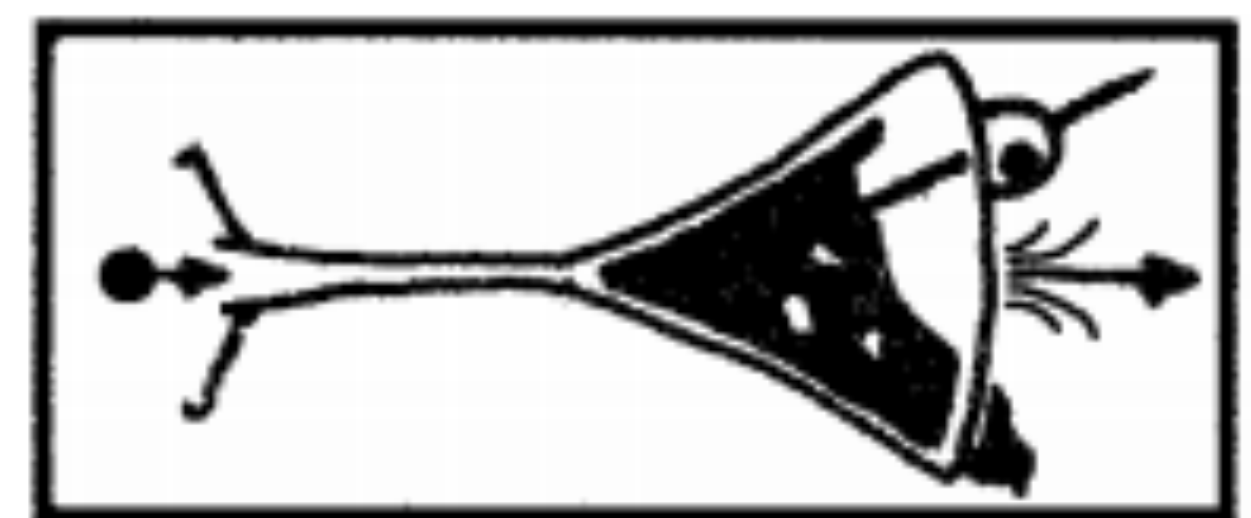
Drill-down story



Interactive slideshow



Martini glass



More
reader
control



More
designer
control

Some **inspirations** from movies...



https://www.youtube.com/watch?v=v4seDVfgwOg&ab_channel=EveryFrameaPainting



Montage / editing

Prescribed order

Where you look explicitly directed

Little reader/viewer agency

Ensemble staging / direction

Less prescribed order

Salience, structure nudges you to look

More reader/viewer agency

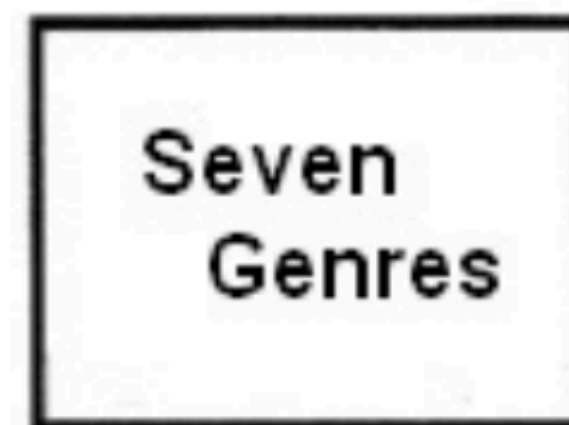
Film uses both, so can we! But the choice should be deliberate



Montage / editing



Ensemble staging / direction



Seven
Genres



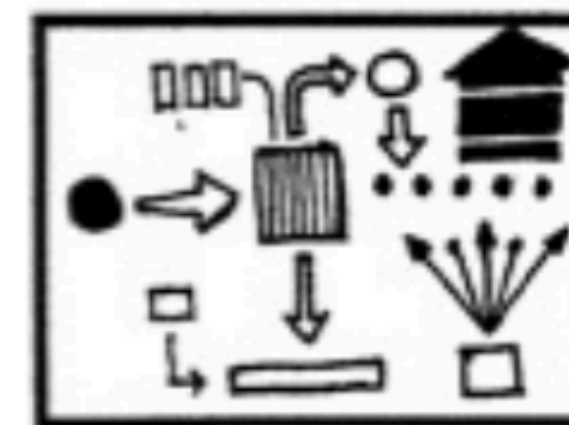
Magazine Style



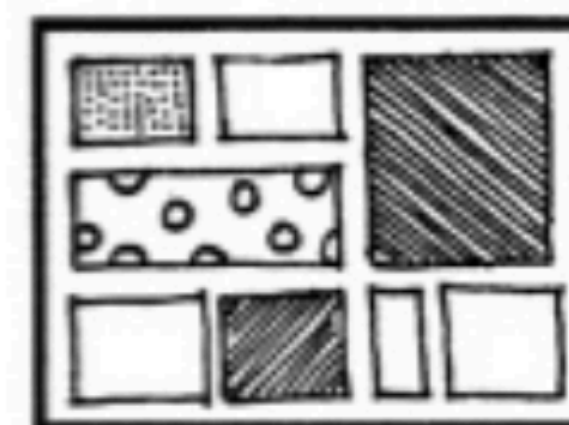
Annotated Chart



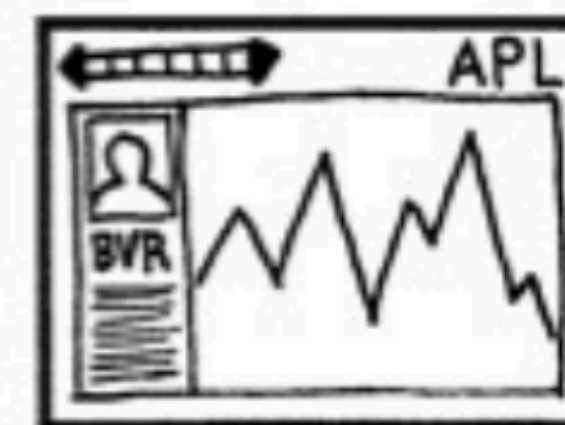
Partitioned Poster



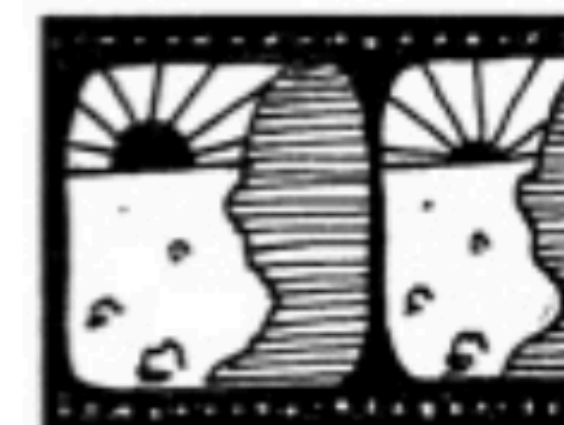
Flow Chart



Comic Strip



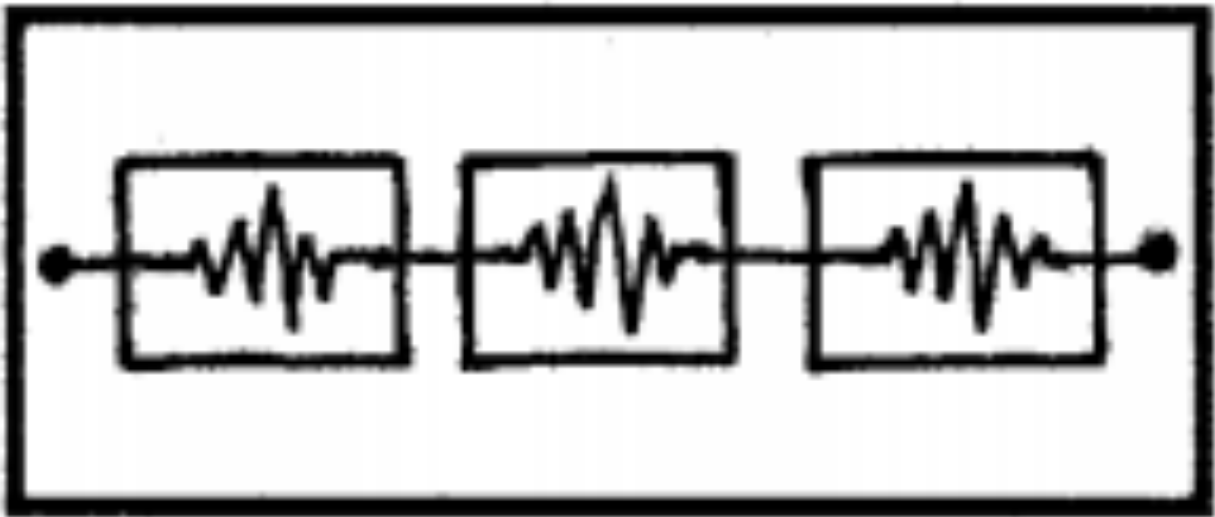
Slide Show



Film/Video/Animation



Montage / editing



Income Mobility Charts for Girls, Asian-Americans and Other Groups. Or Make Your Own.

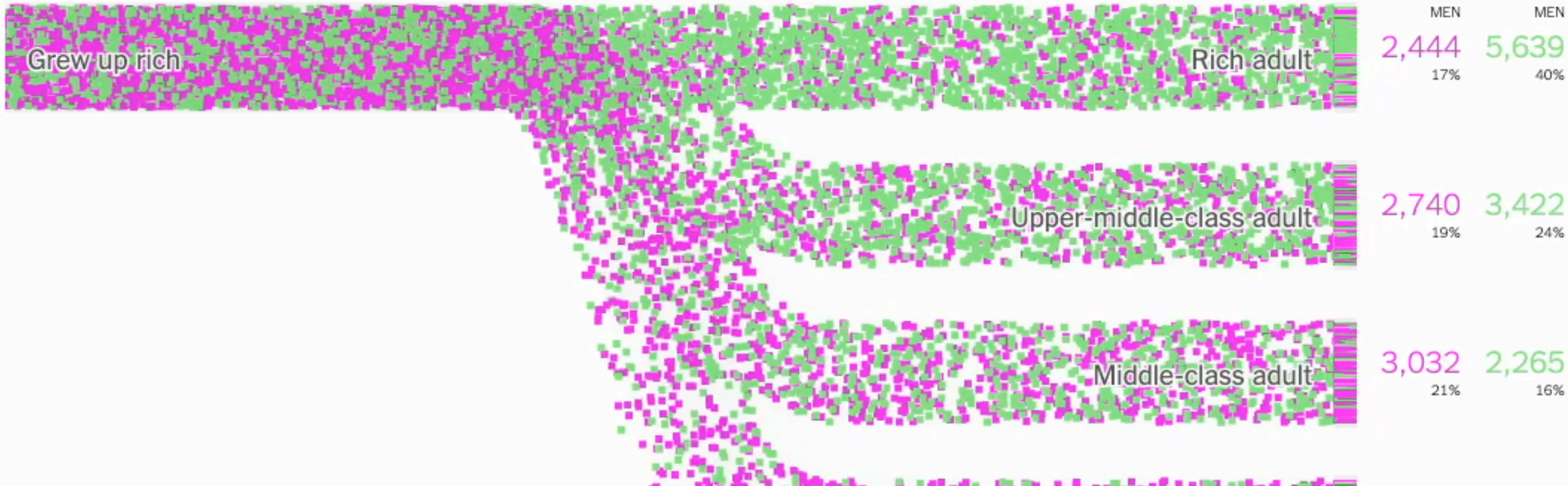
By EMILY BADGER, CLAIRE CAIN MILLER, ADAM PEARCE and KEVIN QUEALY MARCH 27, 2018

Last week we [wrote](#) about a sweeping new study of income inequality, which followed 20 million children in the United States and showed how their adult incomes varied by race and gender. The research was based on data about virtually all Americans now in their late 30s.

This first animated chart illustrates one of the study's main findings: White boys who grow up rich are likely to remain that way, while black boys raised in similarly wealthy households are more likely to fall to the bottom than stay at the top in their own adult households.

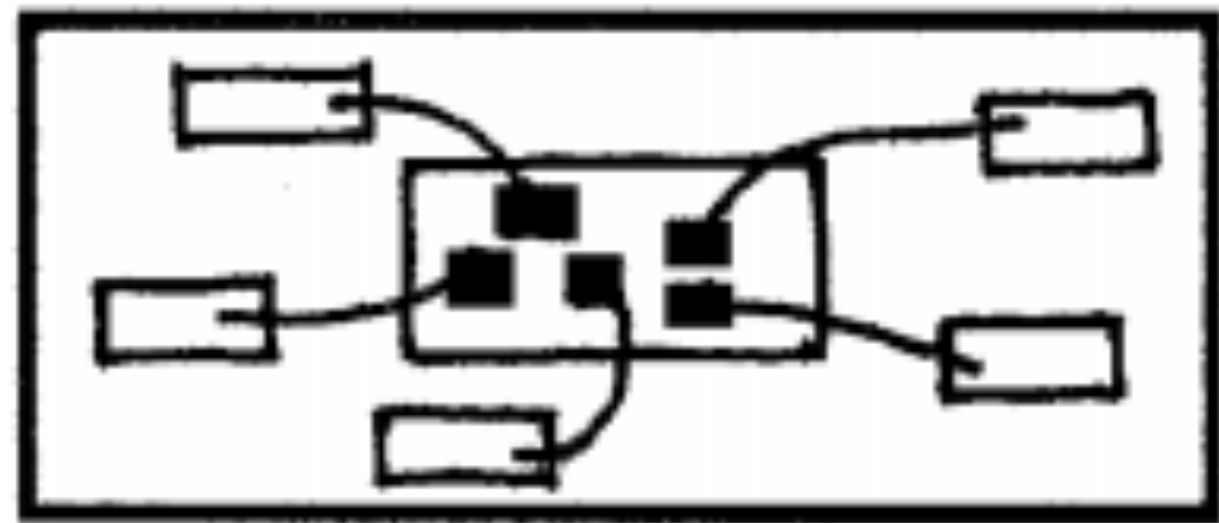
Black and **white** boys raised in wealthy families

Follow the lives of these **41,877** Americans and see where they end up as adults:



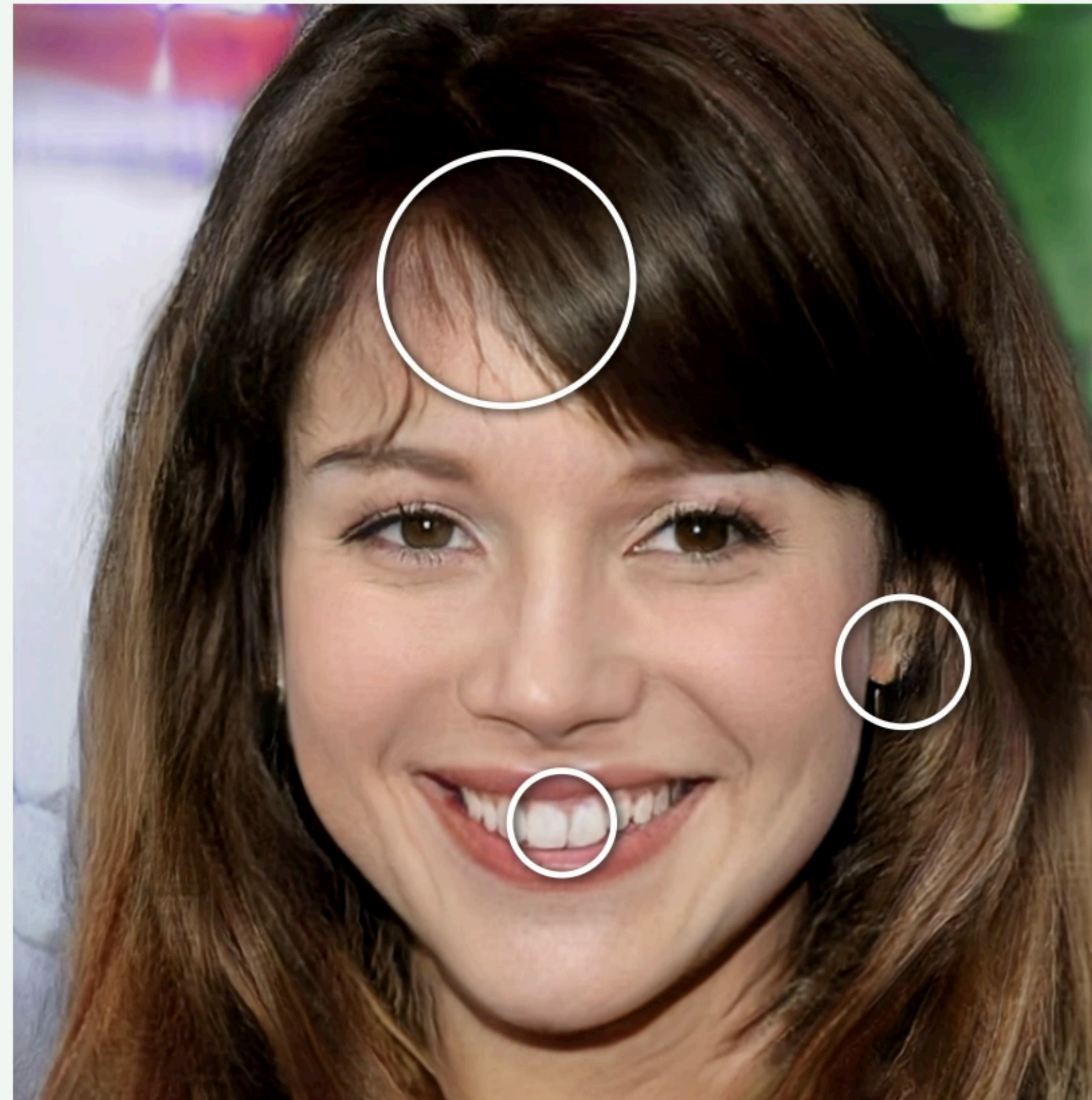


Ensemble staging / direction

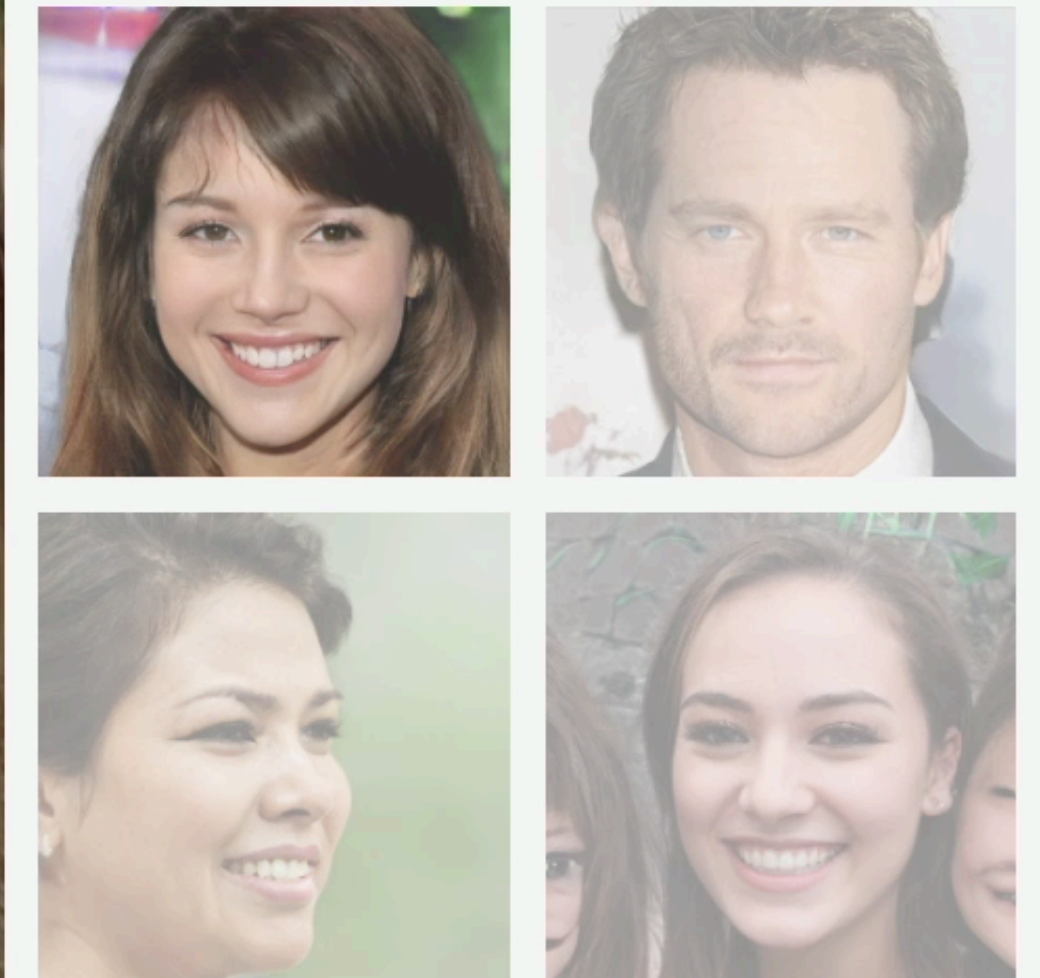


What gives away a machine-generated image?

Interactivity on illustrations can help people get more context around certain objects that may not have clear and separable boundaries.



Select region for more information.



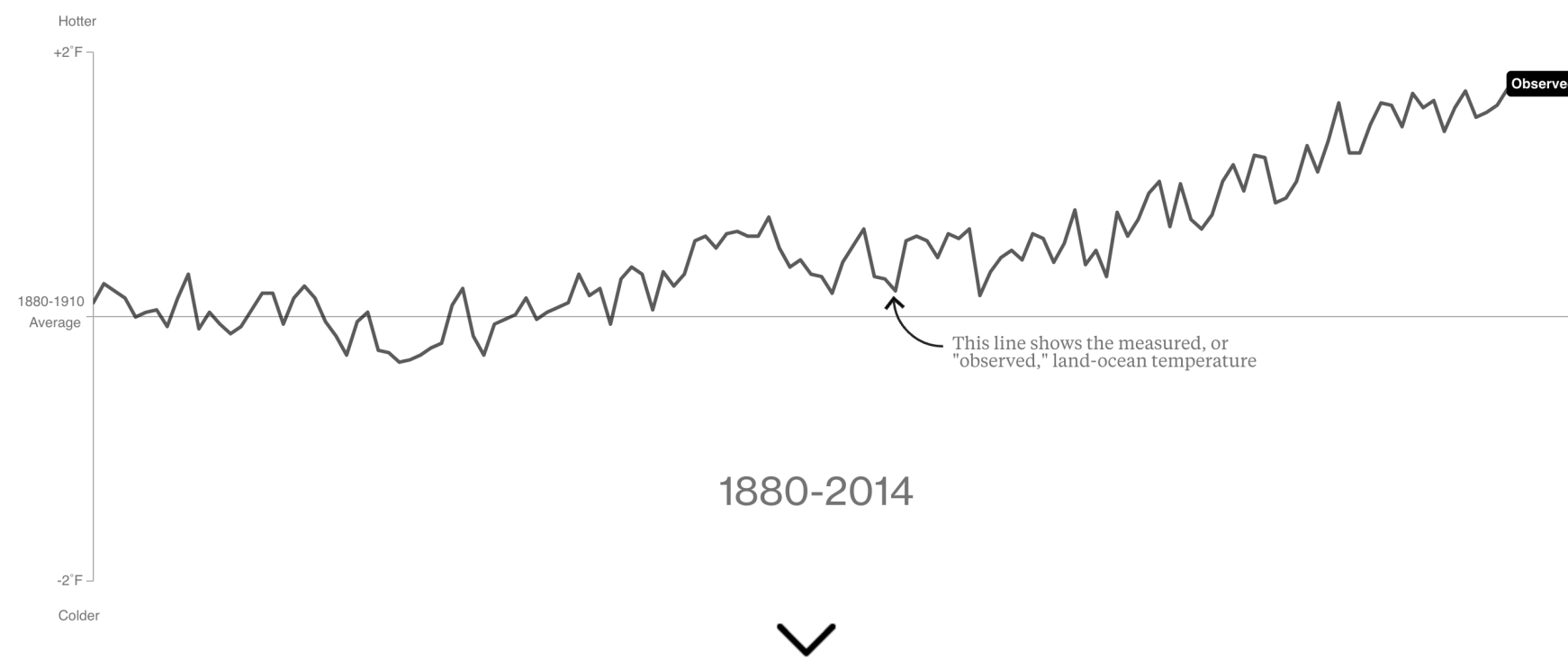
Best if you experience
them yourself...

Activity

What's Really Warming the World?

By Eric Roston and Blacki Migliozi | June 24, 2015

Skeptics of manmade climate change offer various natural causes to explain why the Earth has warmed 1.4 degrees Fahrenheit since 1880. But can these account for the planet's rising temperature? Scroll down to see how much different factors, both natural and industrial, contribute to global warming, based on findings from NASA's Goddard Institute for Space Studies.



<https://www.bloomberg.com/graphics/2015-whats-warming-the-world>

What techniques?

How are they directing attention?

Does the story flow?

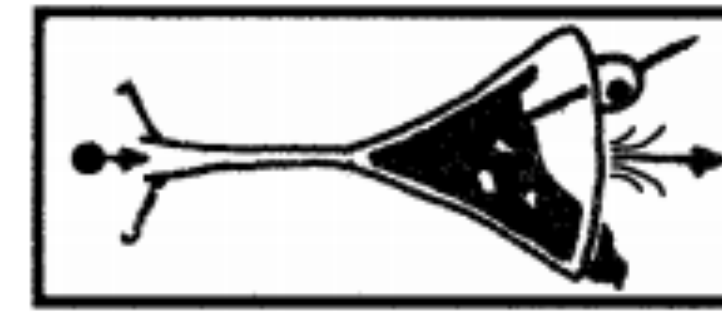
Is it effective / engaging?



Montage / editing



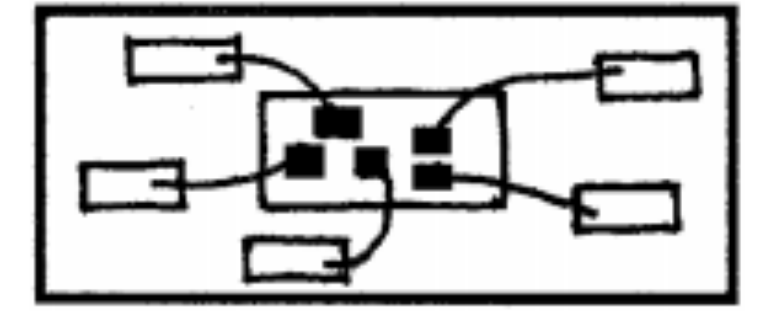
Ensemble staging / direction



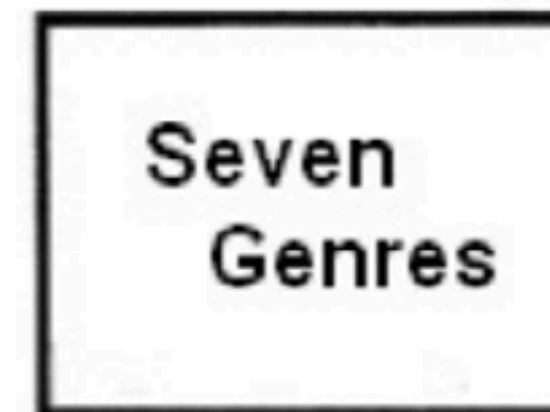
Martini glass



Interactive slideshow



Drill-down story



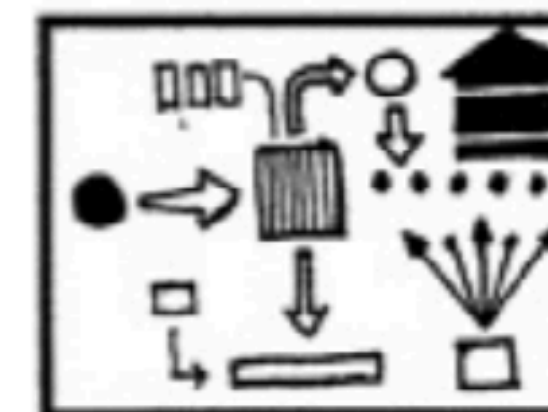
Magazine Style



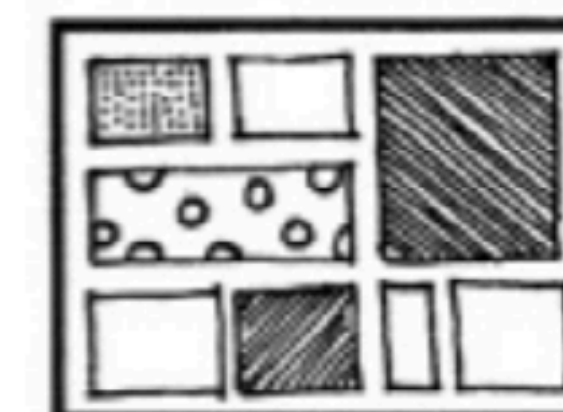
Annotated Chart



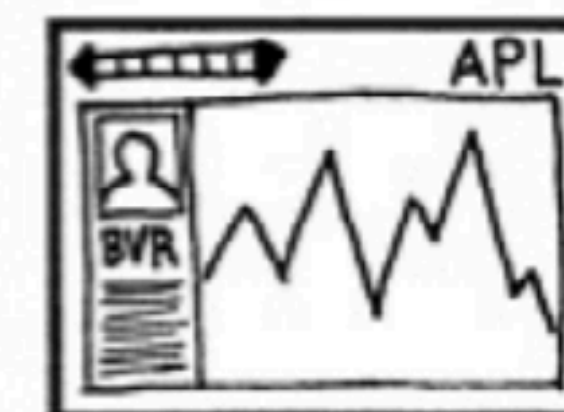
Partitioned Poster



Flow Chart



Comic Strip



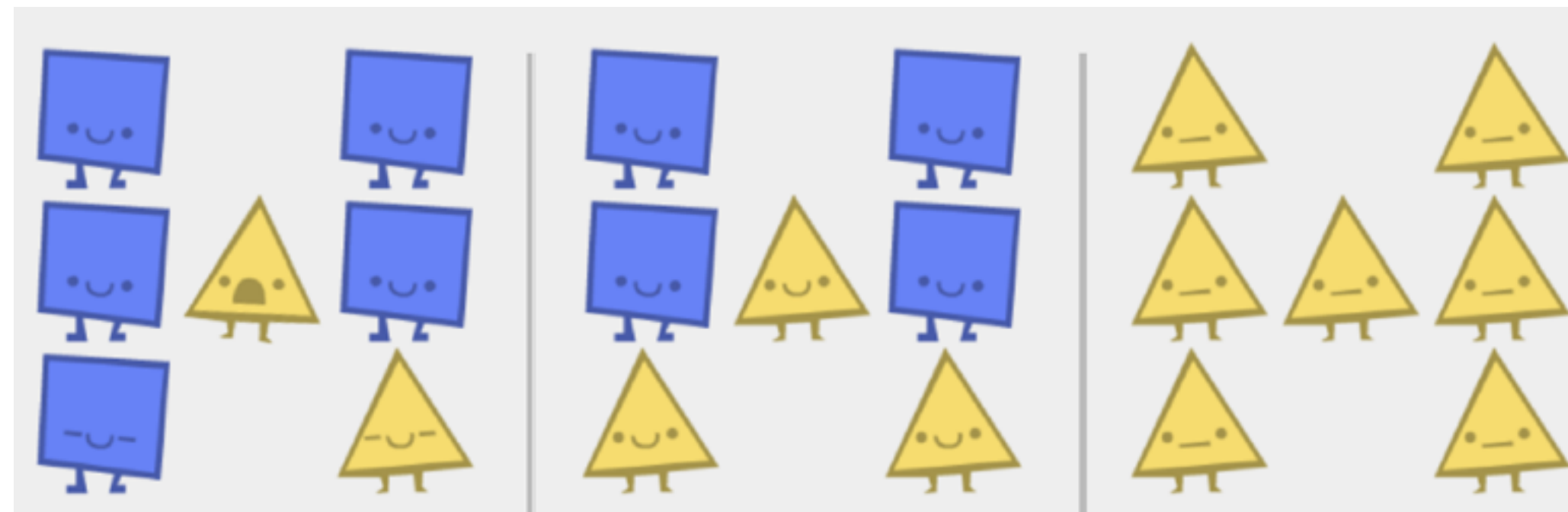
Slide Show



Film/Video/Animation

Activity

Parable of Polygons



<https://ncase.me/polygons/>

What techniques?

How are they directing attention?

Does the story flow?

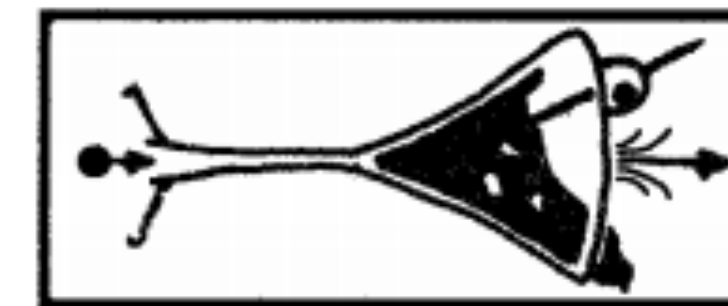
Is it effective / engaging?



Montage / editing



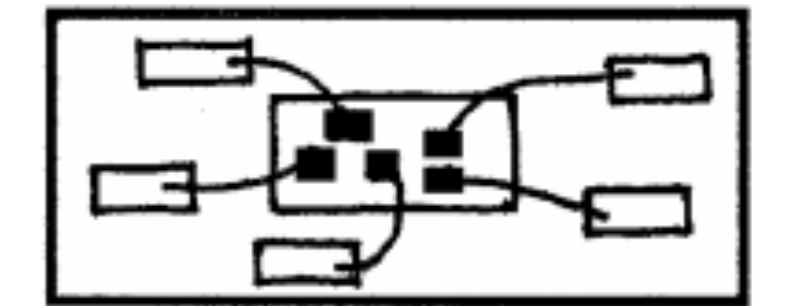
Ensemble staging / direction



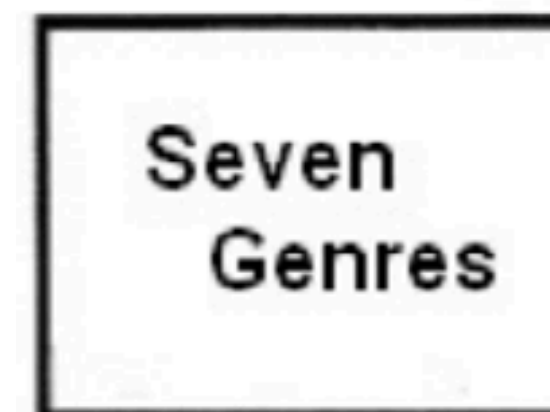
Martini glass



Interactive slideshow



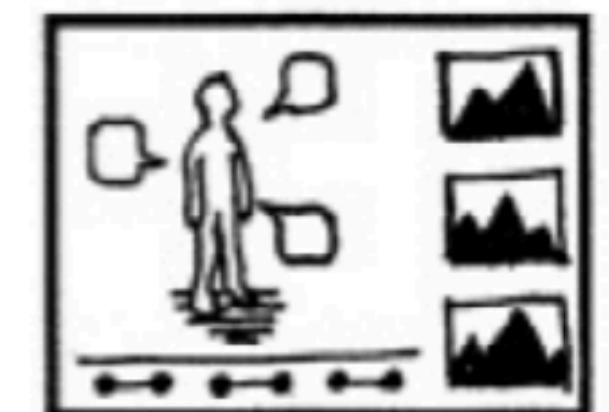
Drill-down story



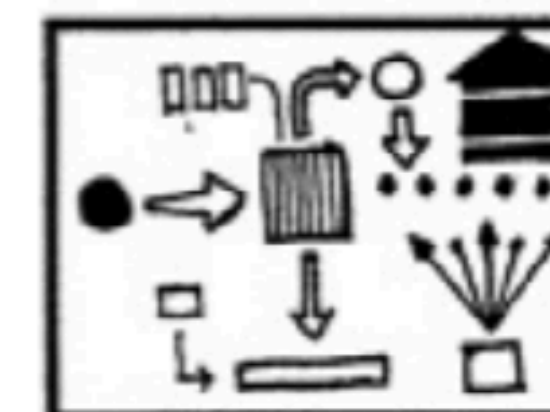
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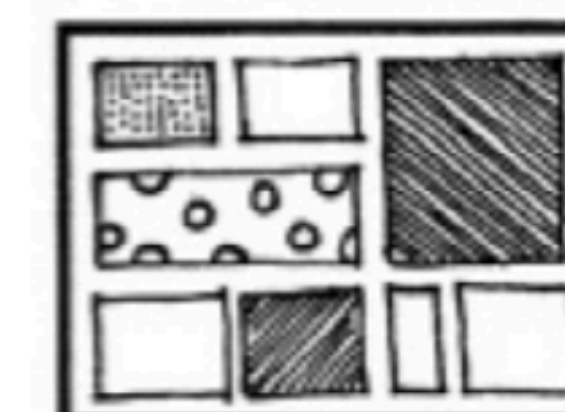
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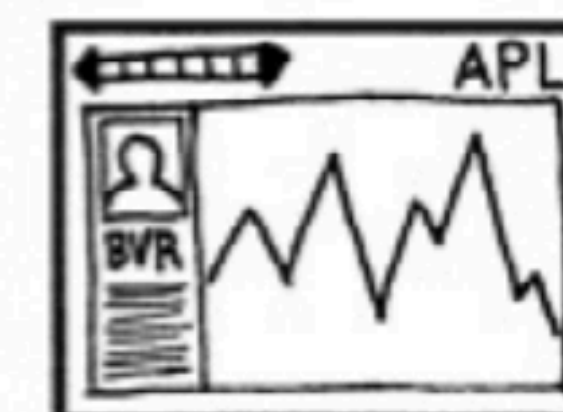
Partitioned Poster



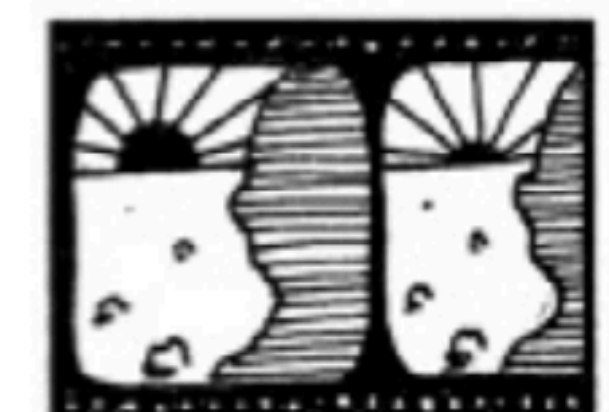
Flow Chart



Comic Strip

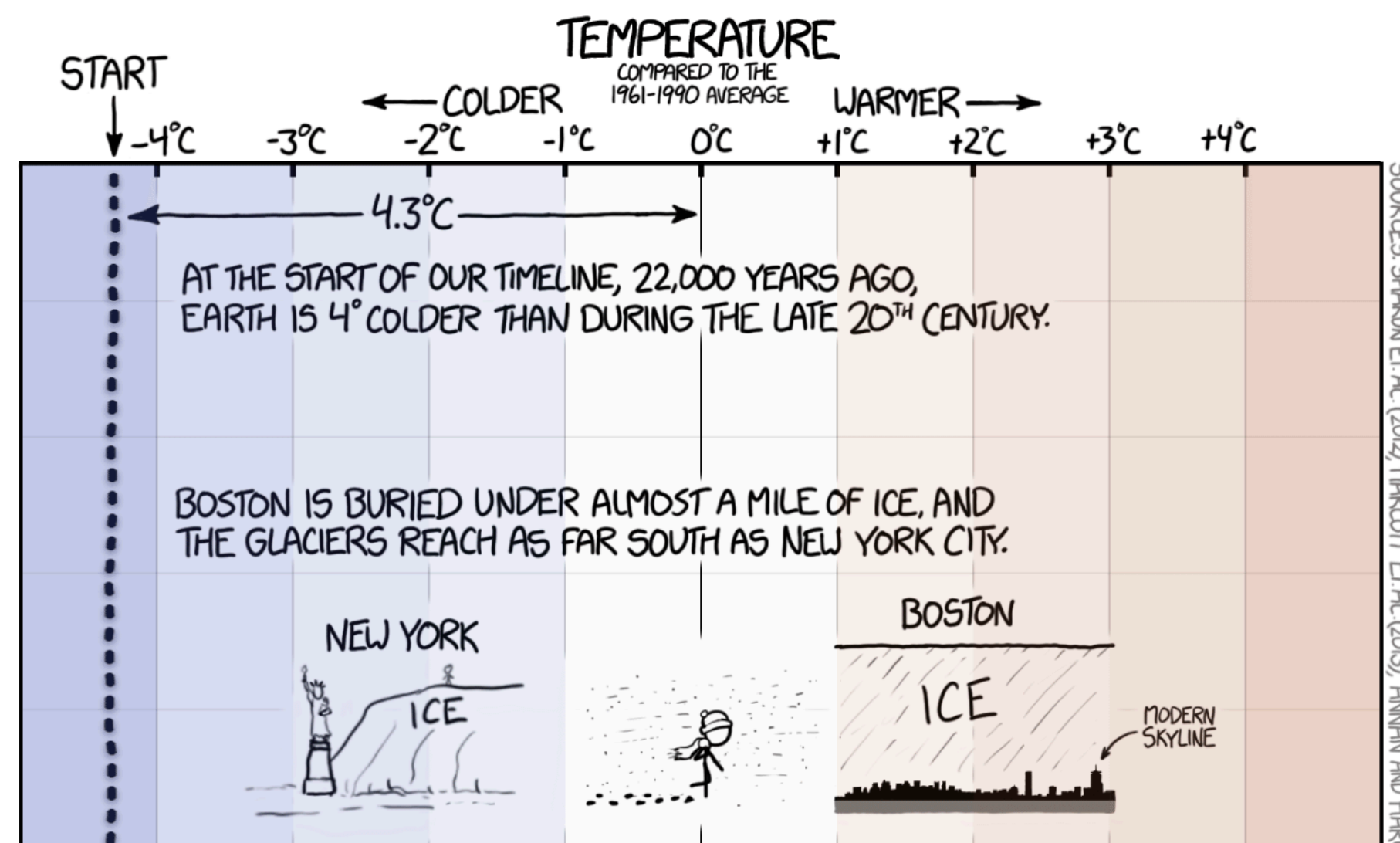


Slide Show



Film/Video/Animation

Activity



<https://xkcd.com/1732/>

What techniques?

How are they directing attention?

Does the story flow?

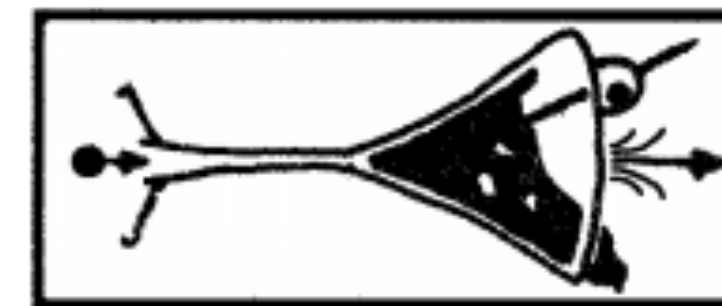
Is it effective / engaging?



Montage / editing



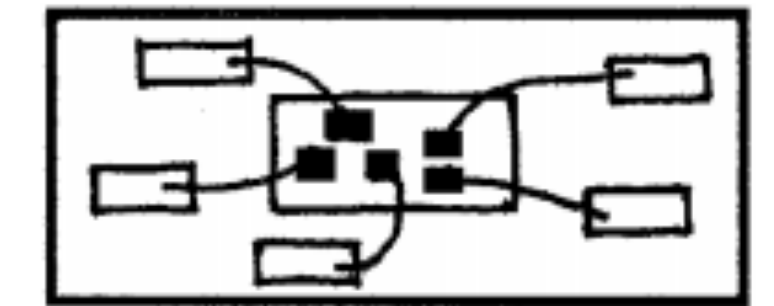
Ensemble staging / direction



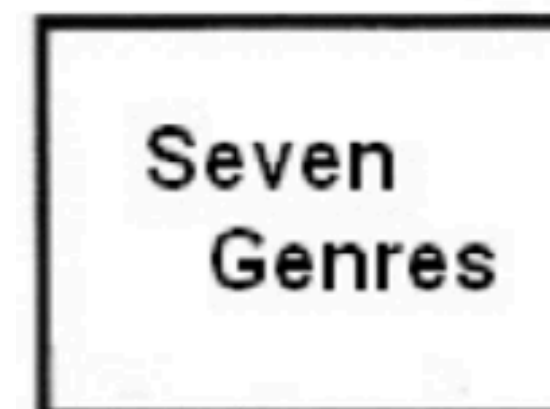
Martini glass



Interactive slideshow



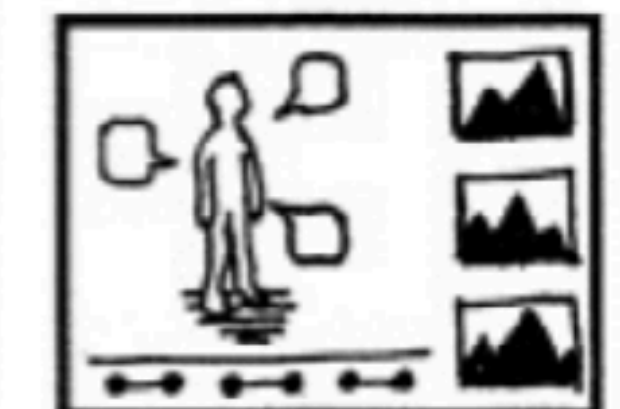
Drill-down story



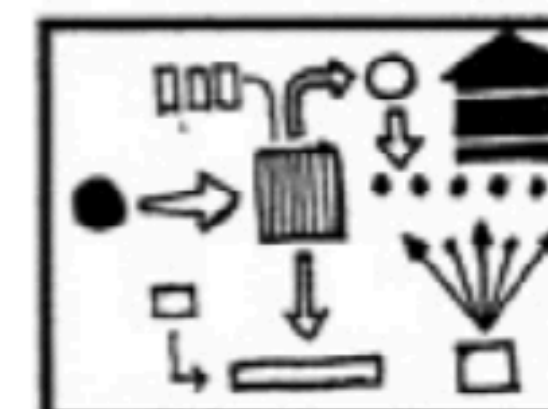
Magazine Style



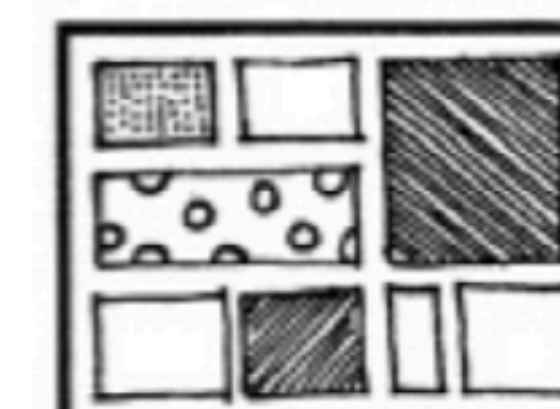
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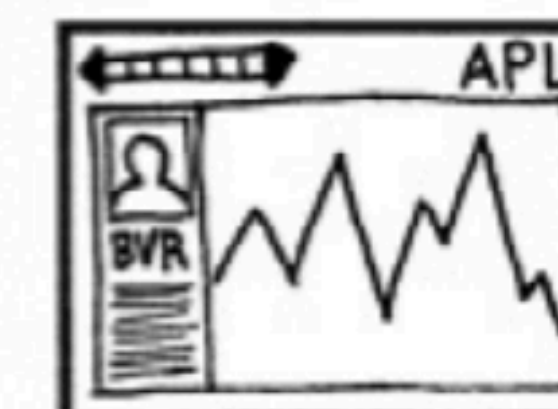
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Flow Chart



Comic Strip



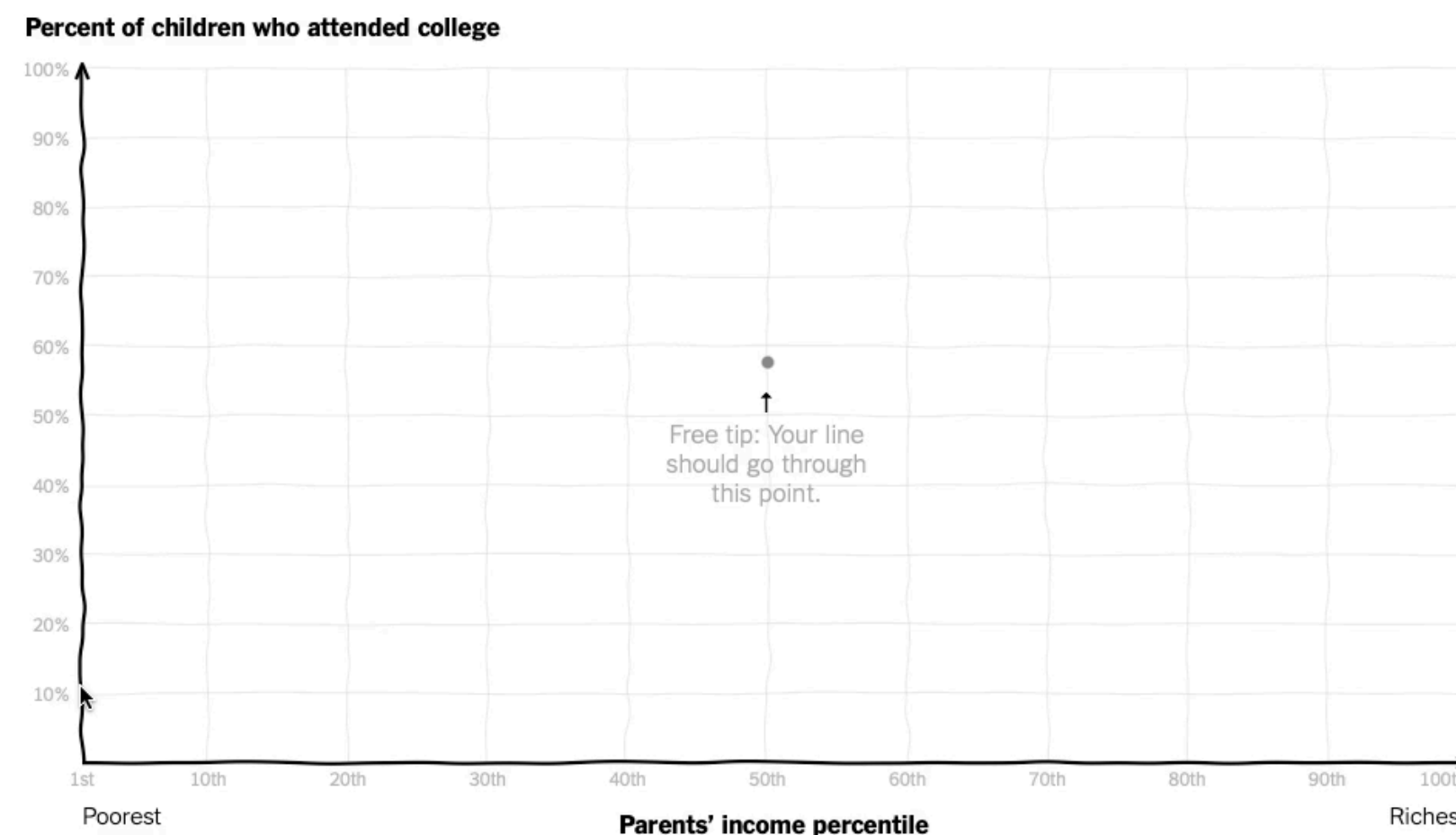
Slide Show



Film/Video/Animation

Activity

Draw your line on the chart below



<https://www.nytimes.com/interactive/2015/05/28/upshot/you-draw-it-how-family-income-affects-childrens-college-chances.html>

What techniques?

How are they directing attention?

Does the story flow?

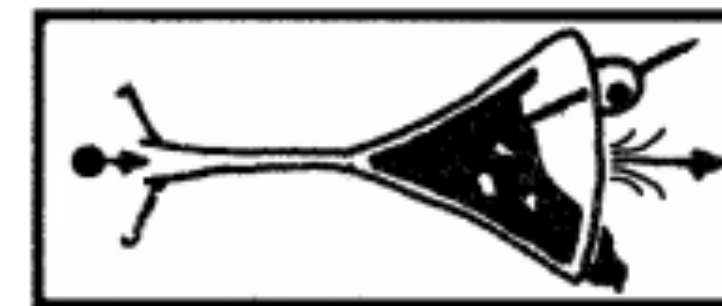
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Montage / editing



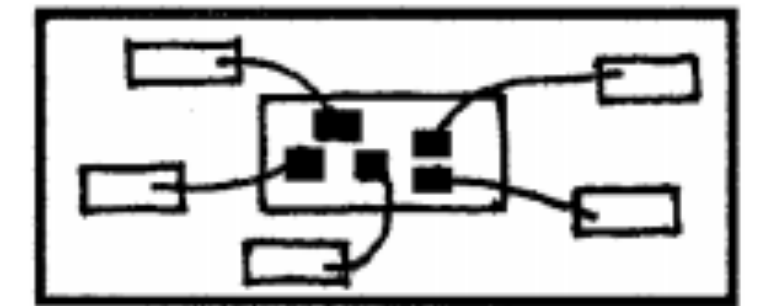
Ensemble staging / direction



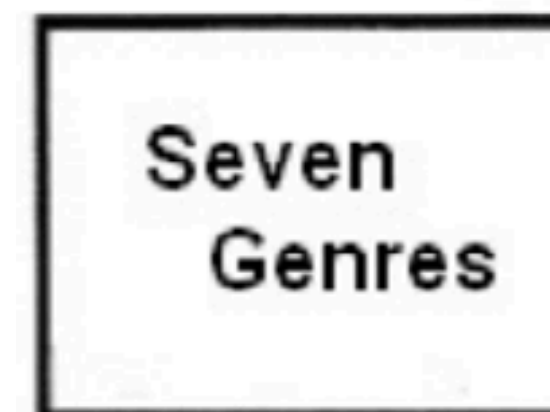
Martini glass



Interactive slideshow



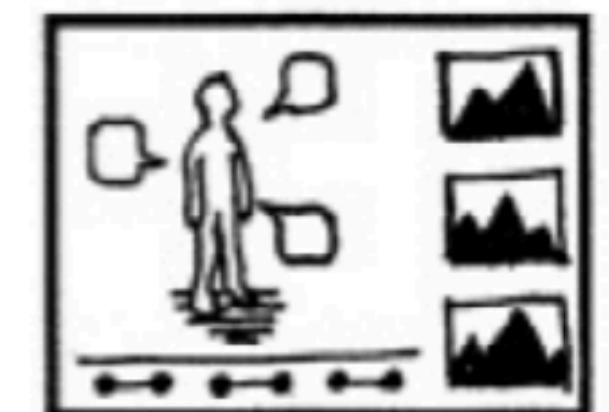
Drill-down story



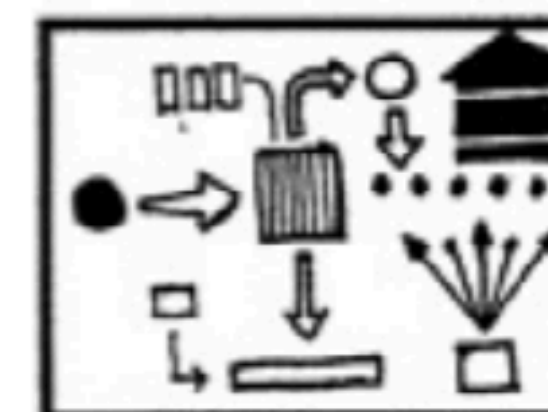
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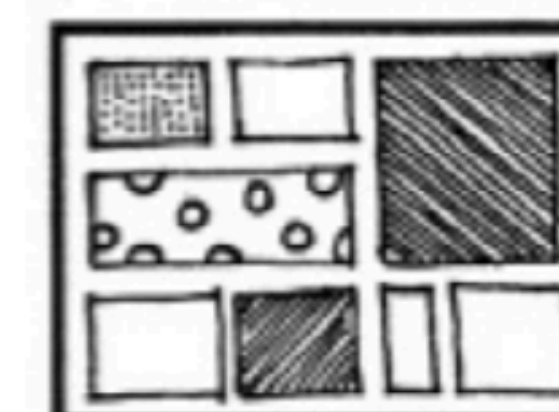
Annotated Chart



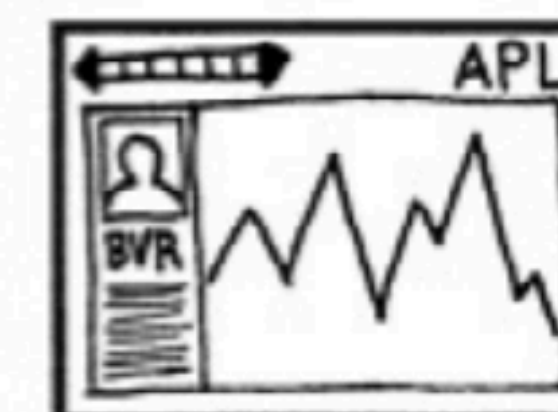
Partitioned Poster



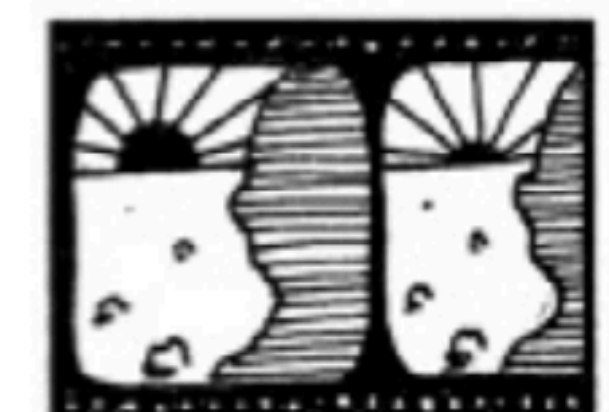
Flow Chart



Comic Strip



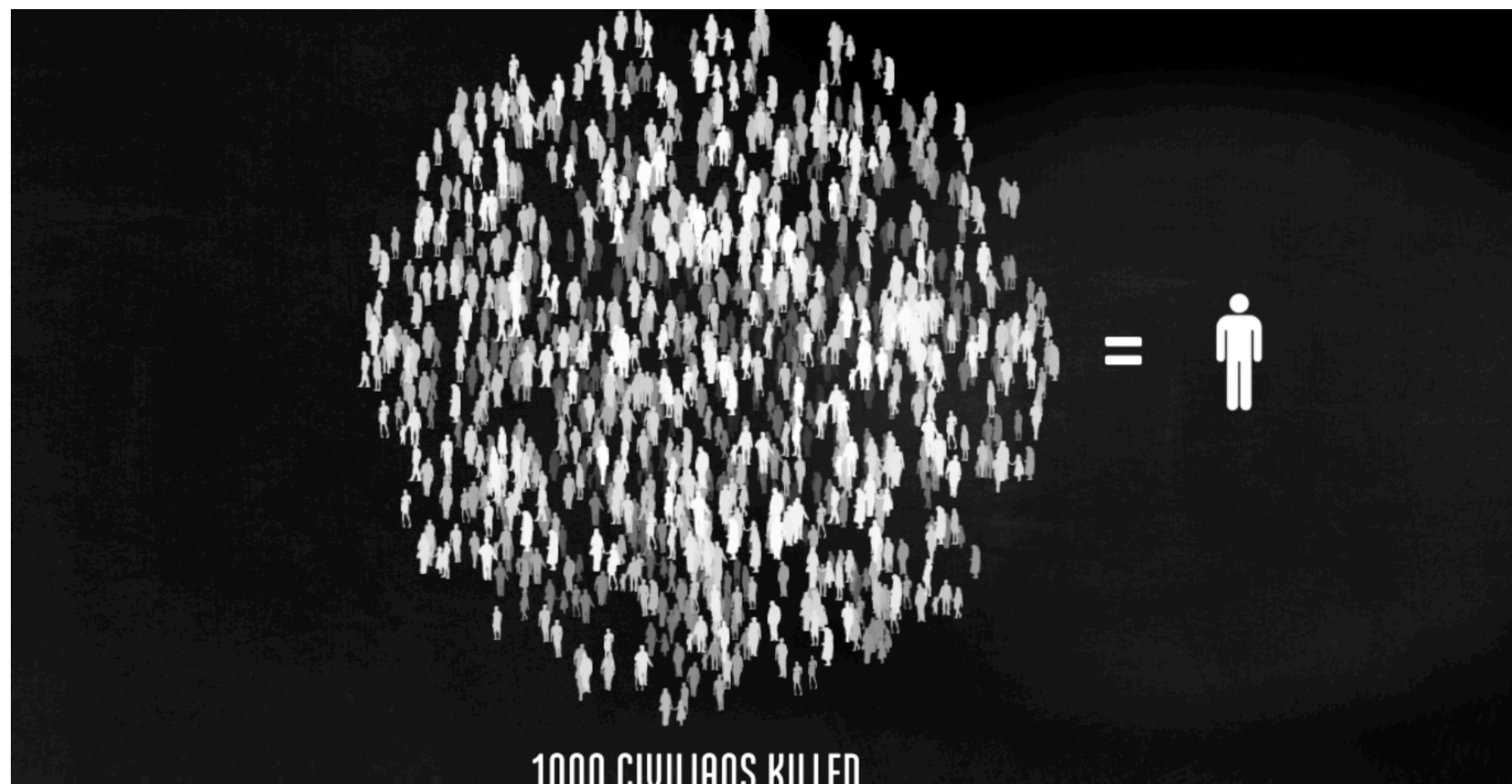
Slide Show



Film/Video/Animation

Activity

The Fallen of World War II



<http://www.fallen.io/ww2/>

What techniques?

How are they directing attention?

Does the story flow?

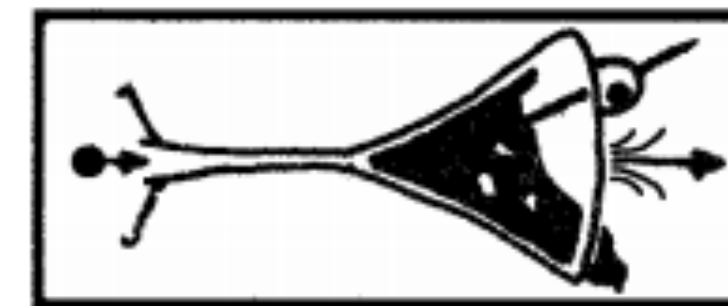
Is it effective / engaging?



Montage / editing



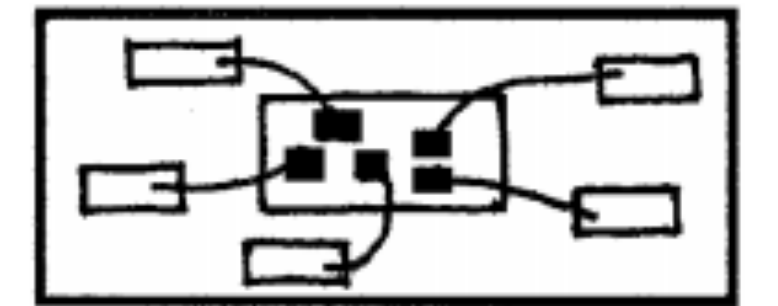
Ensemble staging / direction



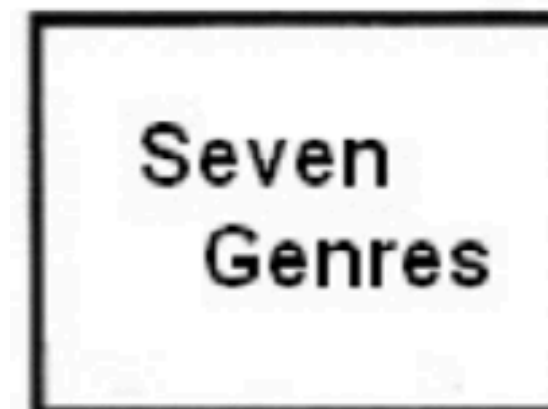
Martini glass



Interactive slideshow



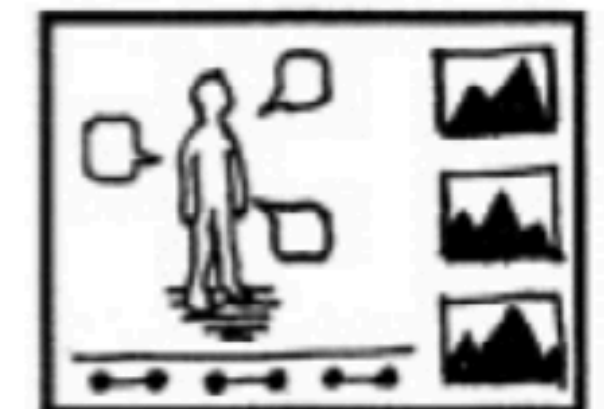
Drill-down story



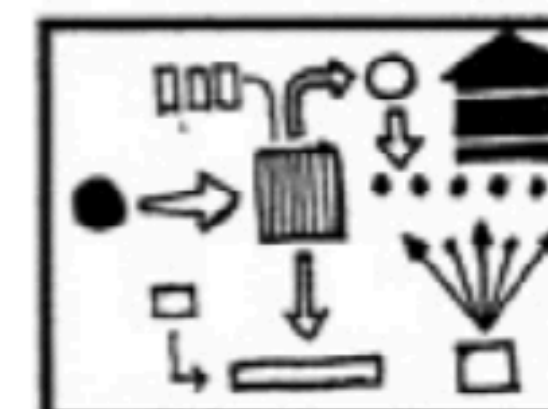
Magazine Style



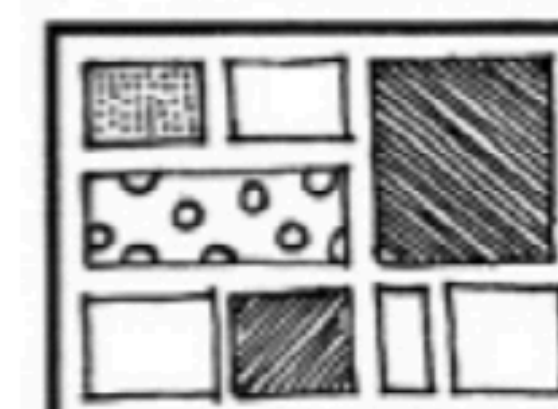
Annotated Chart



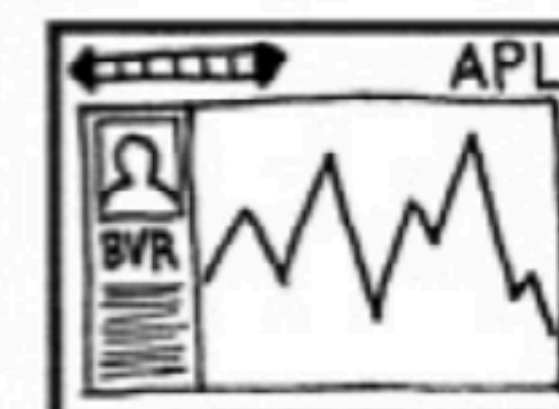
Partitioned Poster



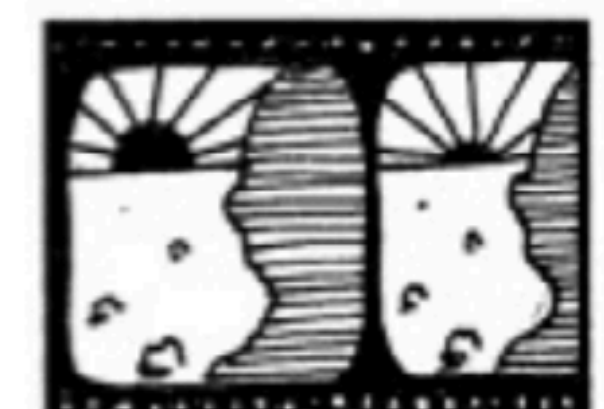
Flow Chart



Comic Strip

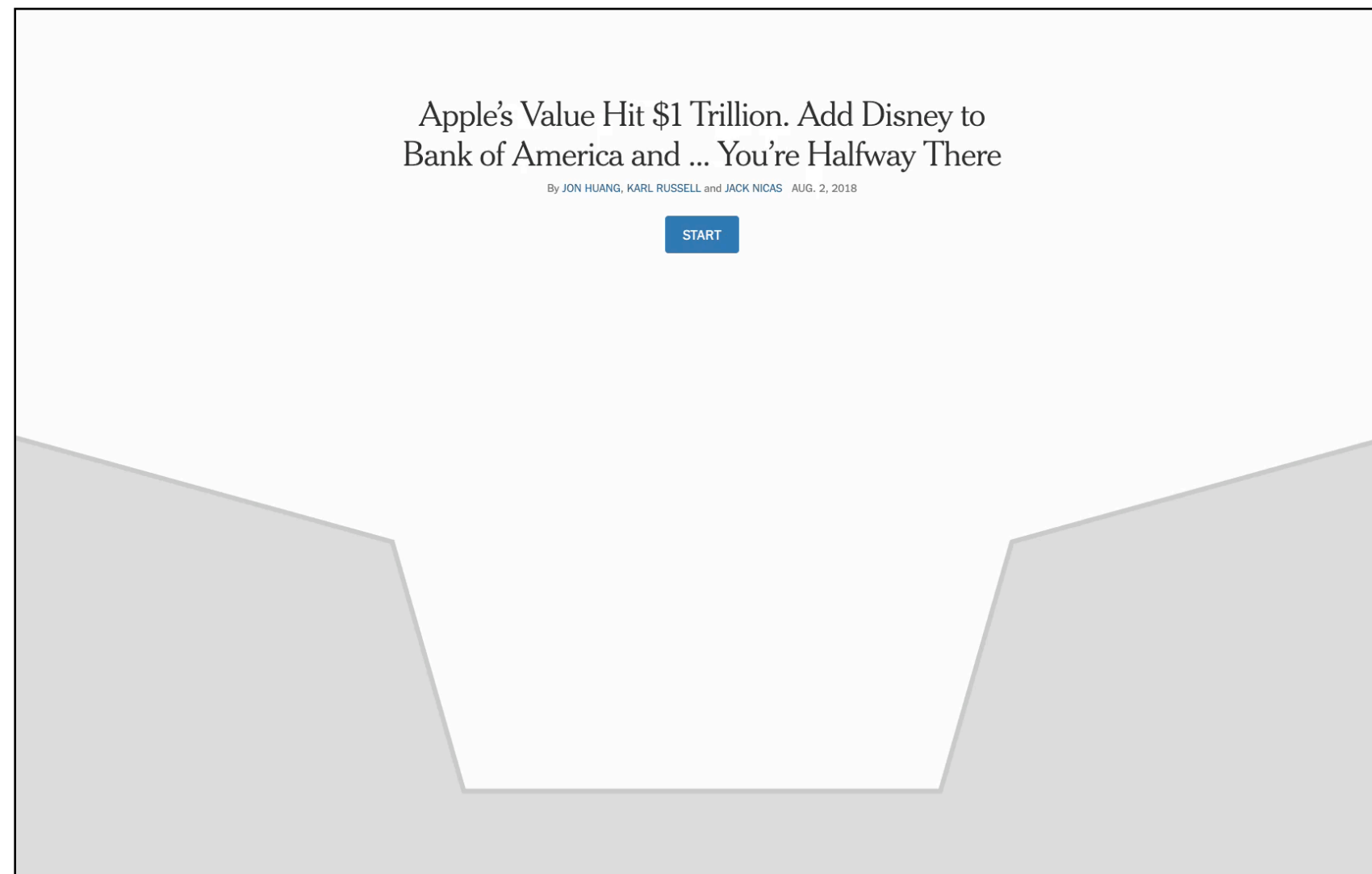


Slide Show



Film/Video/Animation

Activity



<https://www.nytimes.com/interactive/2018/08/02/technology/apple-trillion-market-cap.html>

What techniques?

How are they directing attention?

Does the story flow?

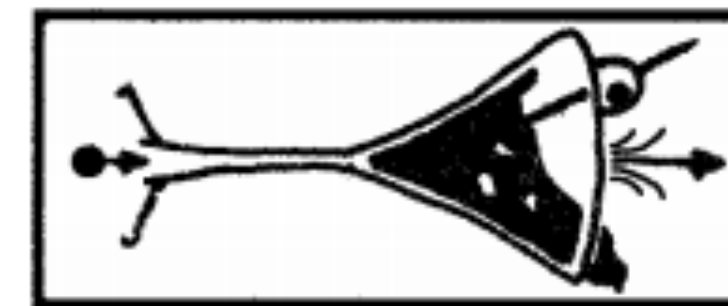
Is it effective / engaging?



Montage / editing



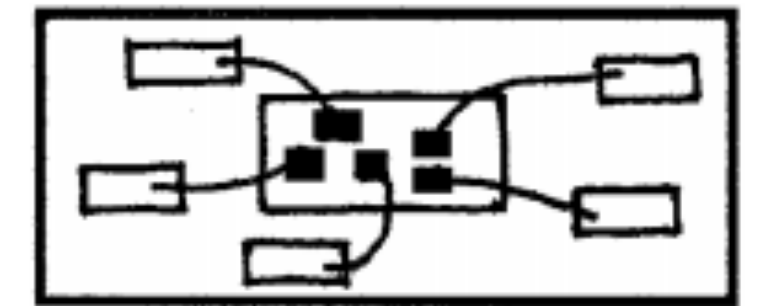
Ensemble staging / direction



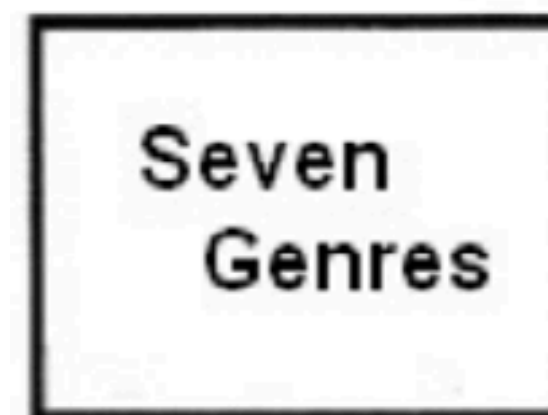
Martini glass



Interactive slideshow



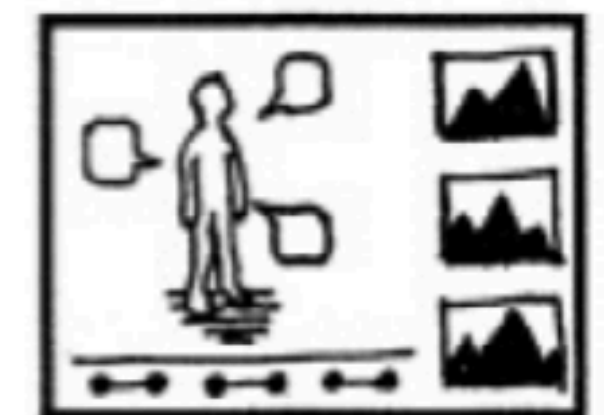
Drill-down story



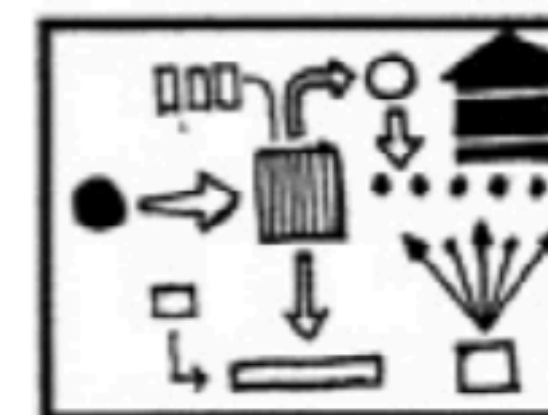
Magazine Style



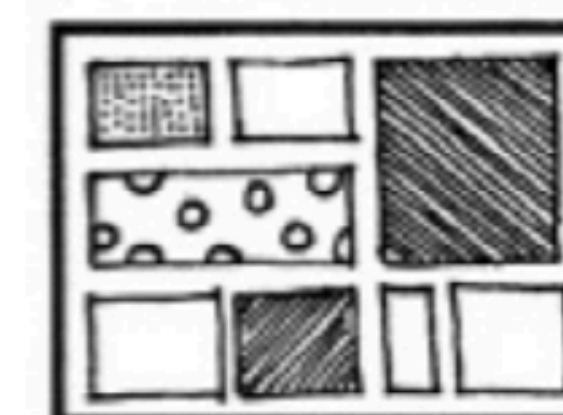
Annotated Chart



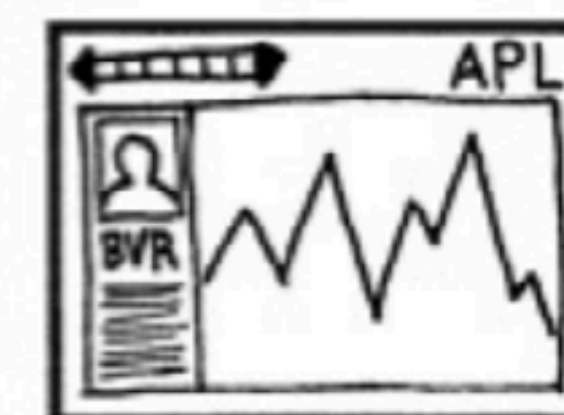
Partitioned Poster



Flow Chart



Comic Strip



Slide Show



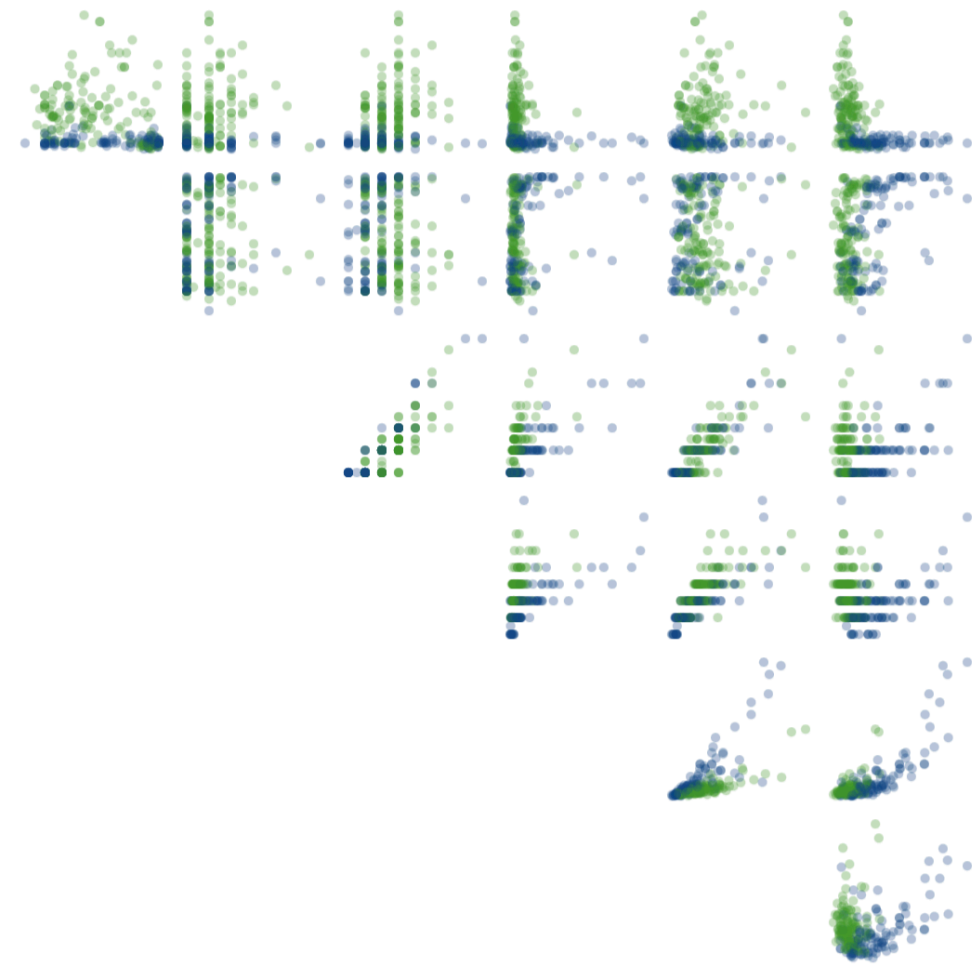
Film/Video/Animation

Activity

A visual introduction to machine learning

English

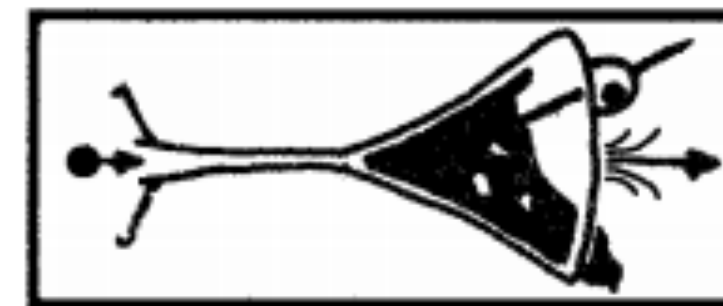
In machine learning, computers apply **statistical learning** techniques to automatically identify patterns in data. These techniques can be used to make highly accurate predictions.



Montage / editing



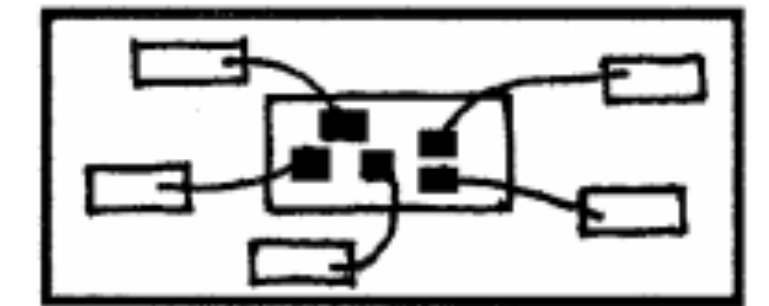
Ensemble staging / direction



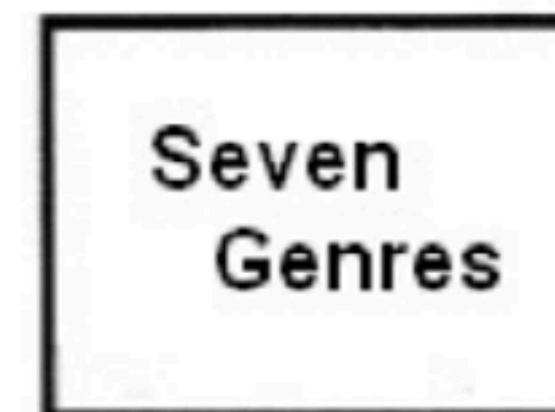
Martini glass



Interactive slideshow



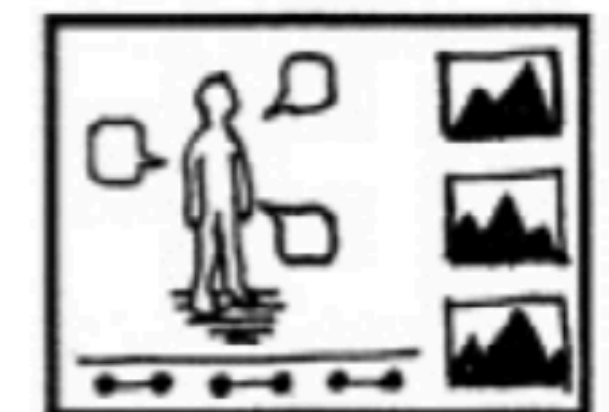
Drill-down story



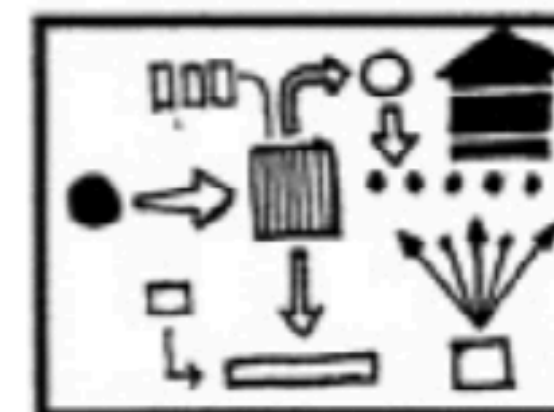
Magazine Style



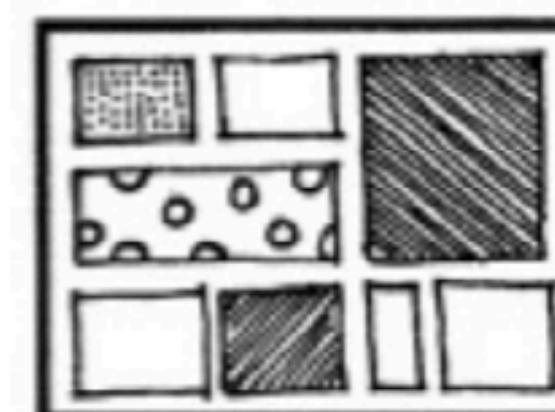
Annotated Chart



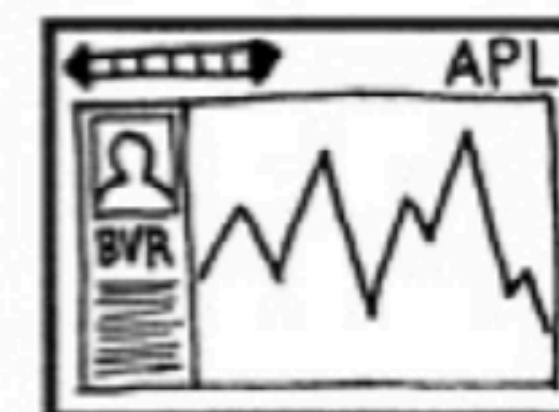
Partitioned Poster



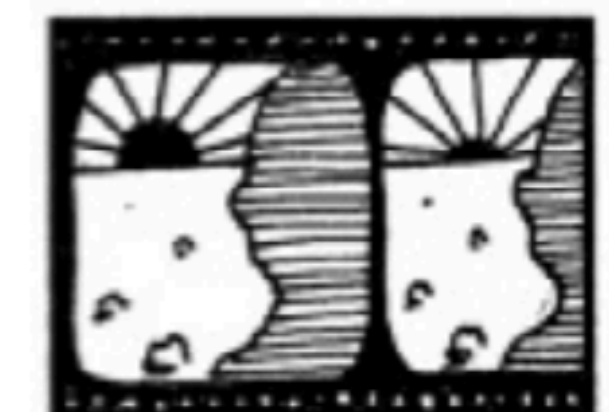
Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

What techniques?

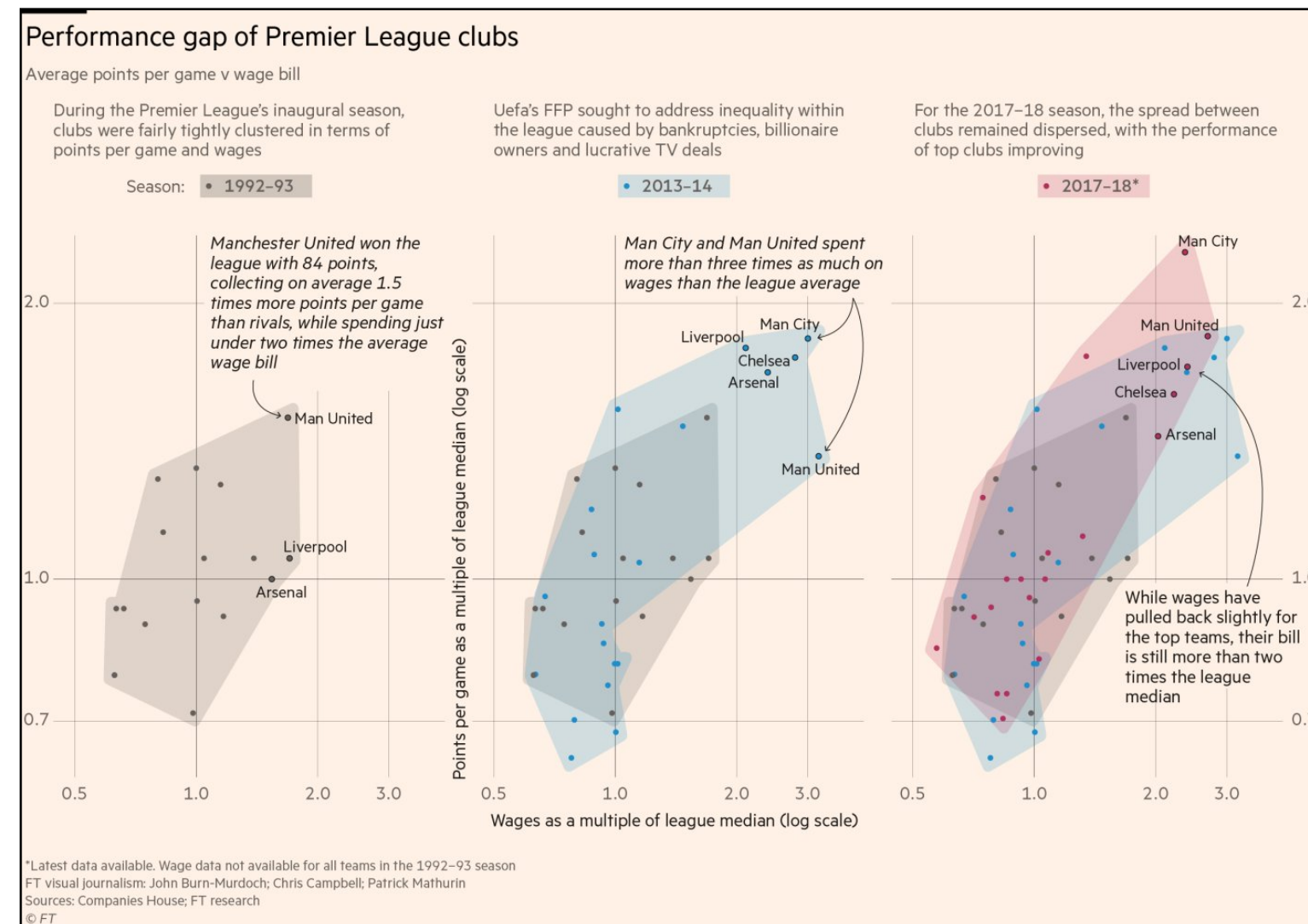
How are they directing attention?

Does the story flow?

Is it effective / engaging?

<https://www.r2d3.us/visual-intro-to-machine-learning-part-1/>

Activity



<https://x.com/ftdata/status/1231157648839651328/photo/1>

What techniques?

How are they directing attention?

Does the story flow?

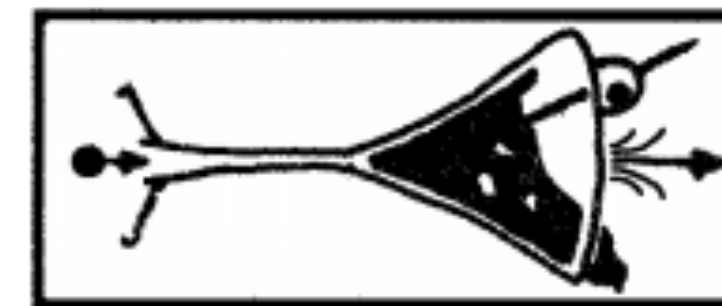
Is it effective / engaging?



Montage / editing



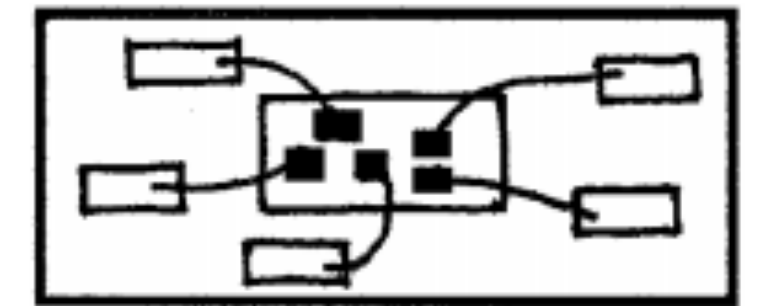
Ensemble staging / direction



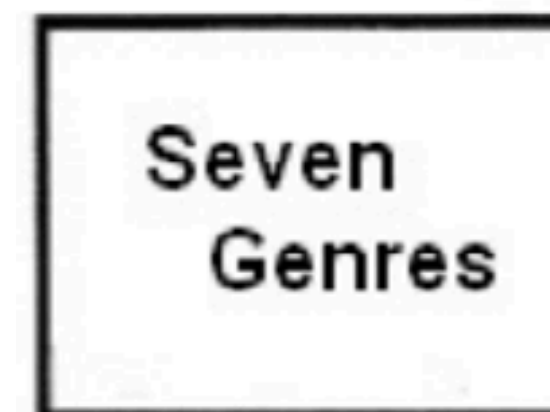
Martini glass



Interactive slideshow



Drill-down story



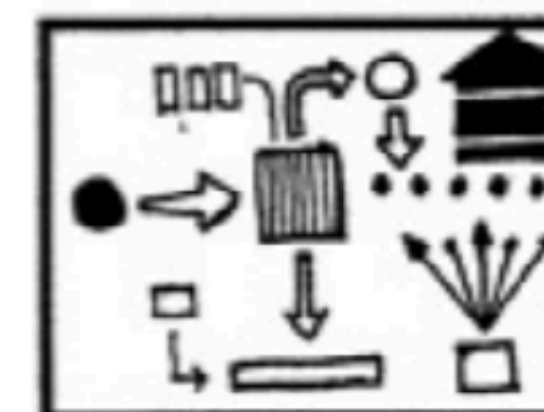
Magazine Style



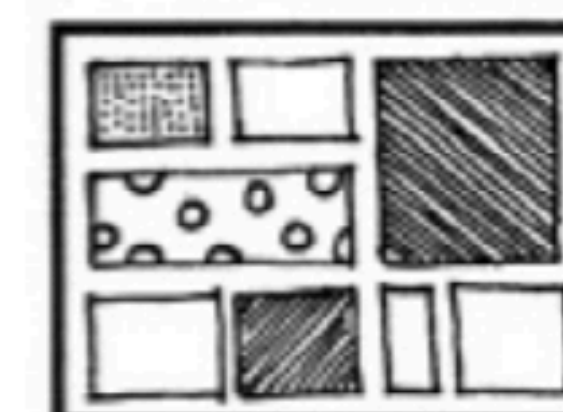
Annotated Chart



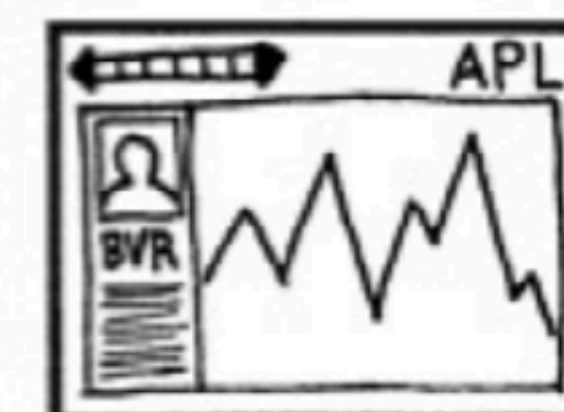
Partitioned Poster



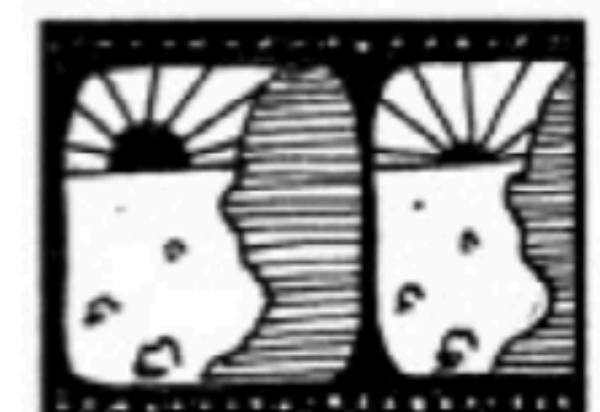
Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

Activity



<https://fivethirtyeight.com/features/gun-deaths/>

What techniques?

How are they directing attention?

Does the story flow?

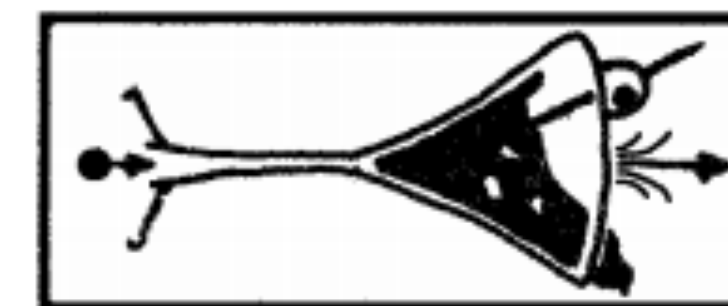
Is it effective / engaging?



Montage / editing



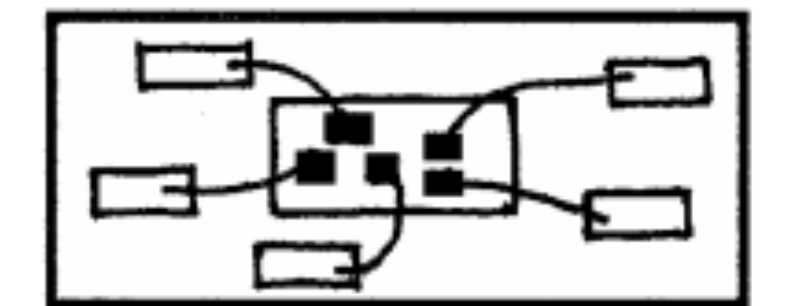
Ensemble staging / direction



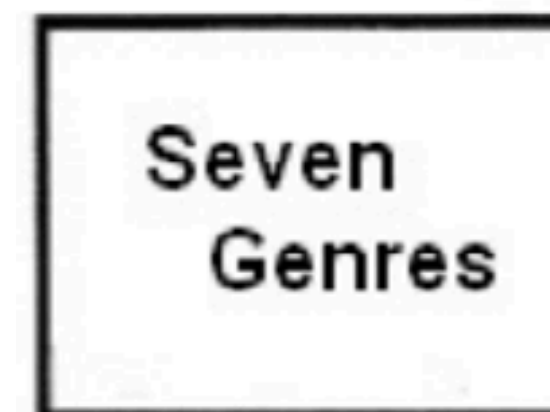
Martini glass



Interactive slideshow



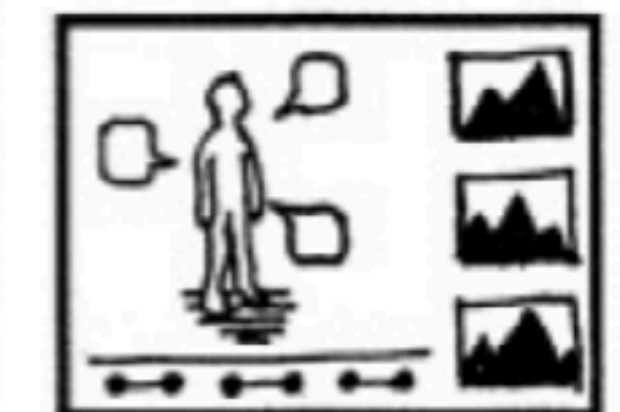
Drill-down story



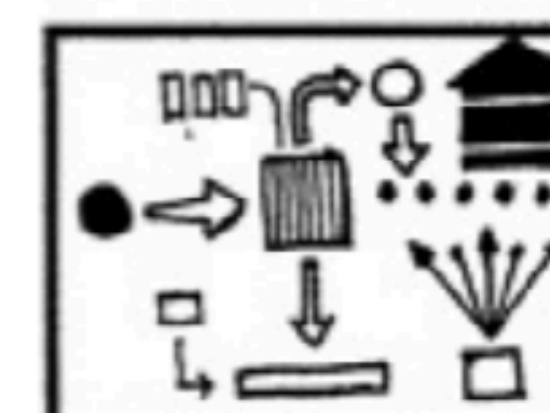
Magazine Style



Annotated Chart



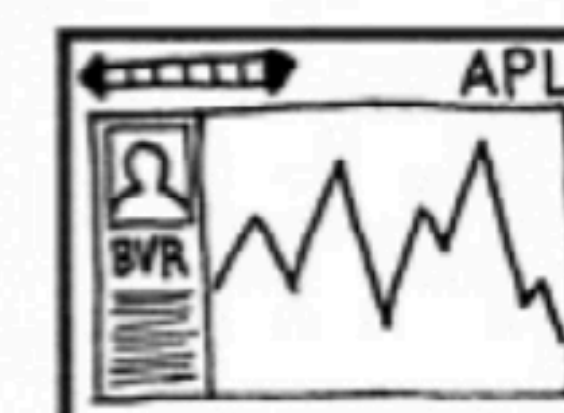
Partitioned Poster



Flow Chart



Comic Strip



Slide Show



Film/Video/Animation

Affordances of the format

Connecting people and data

Making systems playful

Promoting self reflection

Personalizing reading

Use text and annotation!

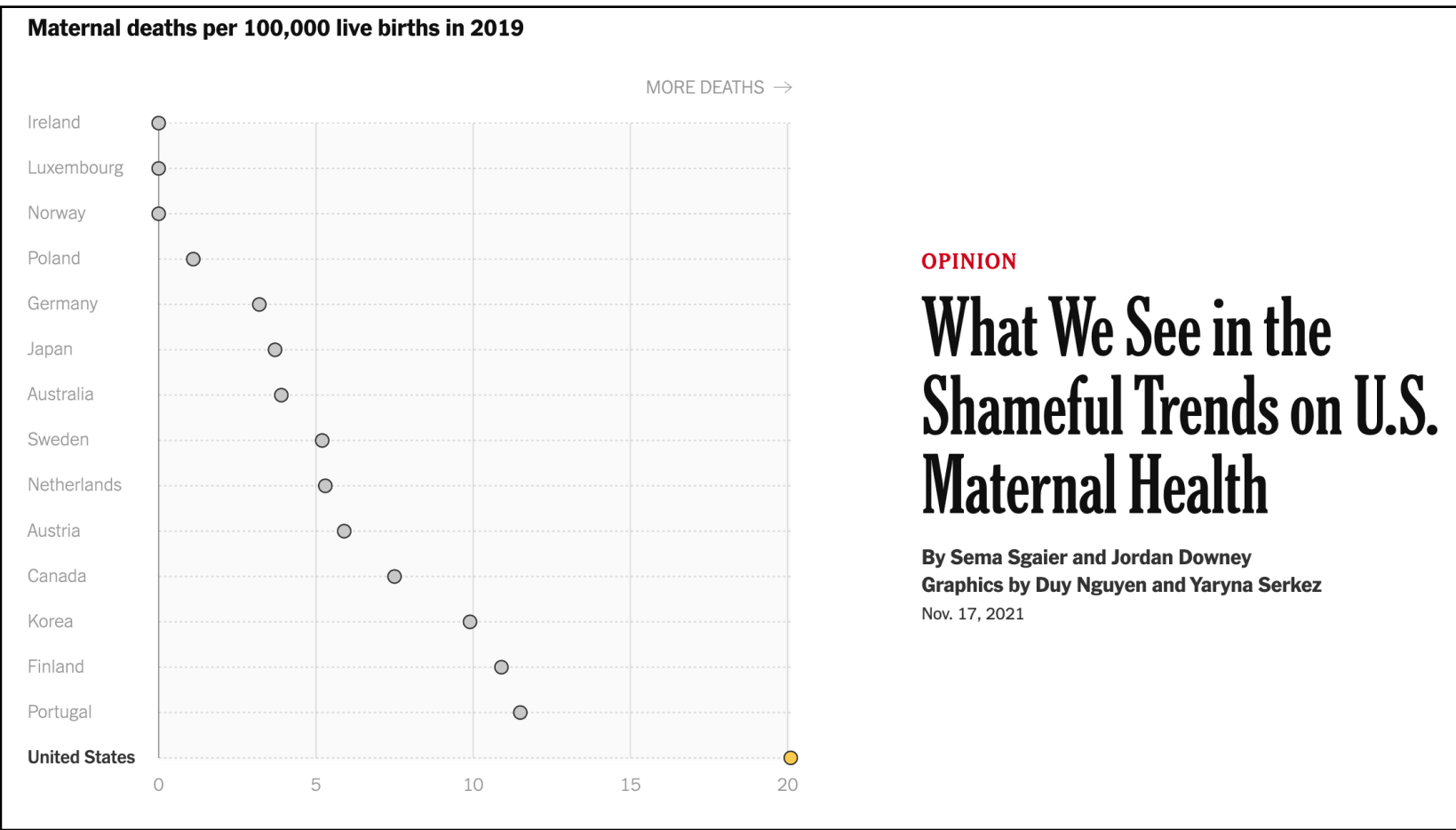
Think about **order** and how **explicit** it needs to be

Tension, suspense, surprise, uncertainty...

Take inspiration from other work, like film, comic, art

*I definitely encourage
to make an engaging
storytelling vis/video
for your final project!*

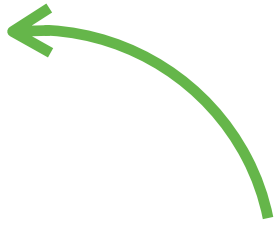
Other examples



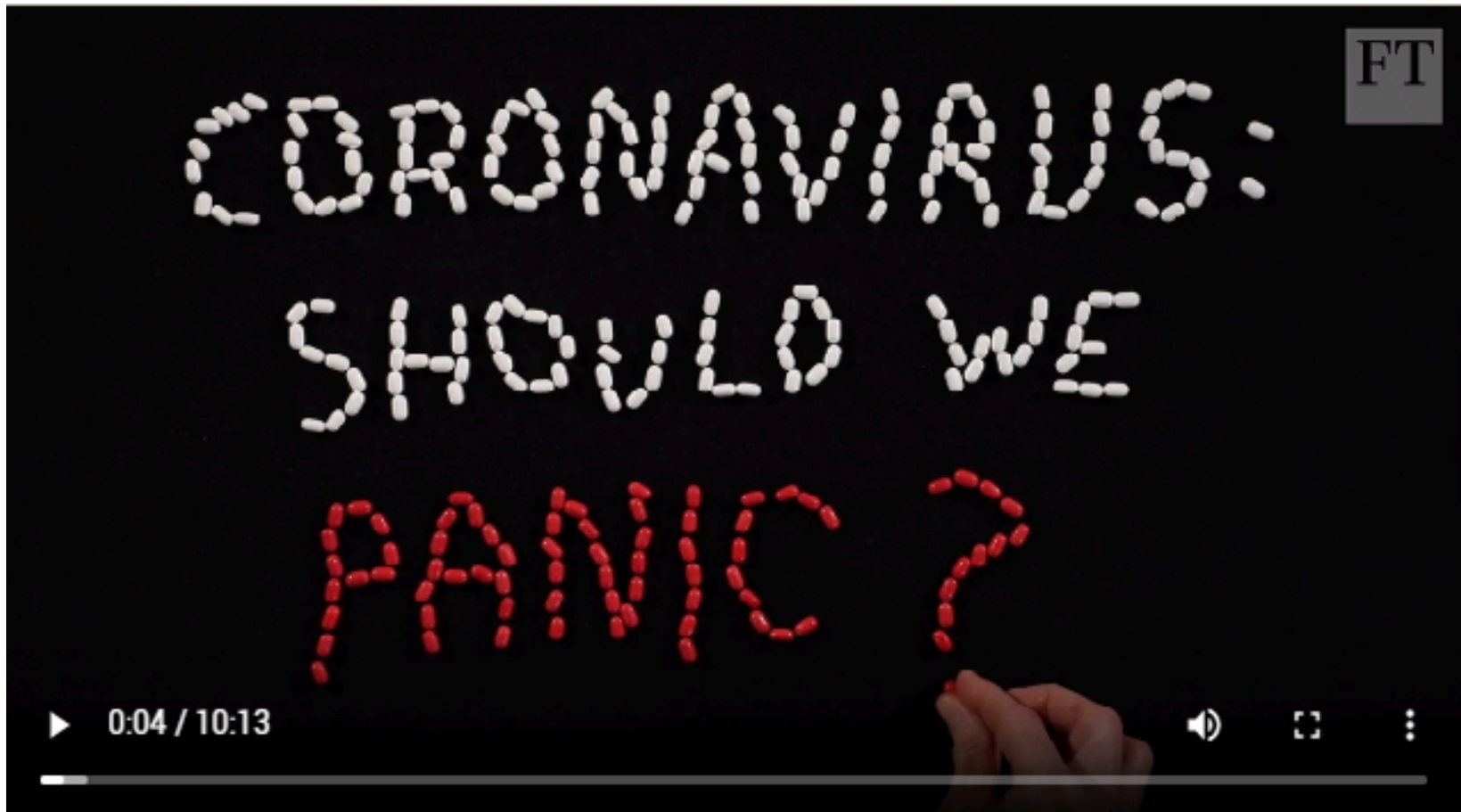
<https://www.nytimes.com/interactive/2021/11/17/opinion/maternal-pregnancy-health.html>



<https://www.nytimes.com/interactive/2020/03/22/world/coronavirus-spread.html>

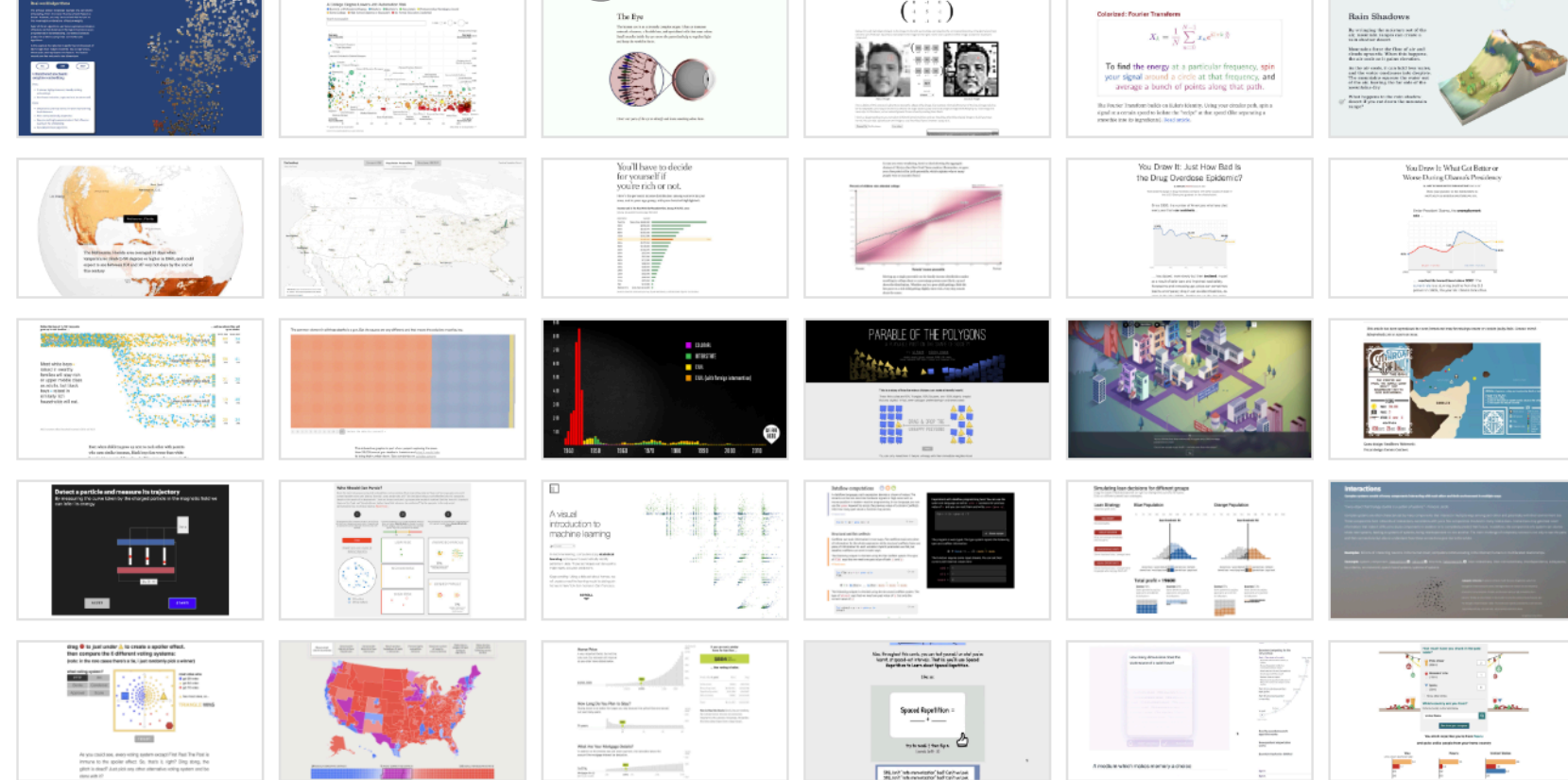


N.B.: This article uses excellent vis and interactivity, but its content is controversial



<https://tinyurl.com/tbesbd4>

Communicating with Interactive Articles



<https://distill.pub/2020/communicating-with-interactive-articles/>

Wed Lab

Faster / desktop: download

<https://terpware.umd.edu/search?q=Tableau%20Installer>

Slower / web-based: sign up

<https://public.tableau.com/app/discover>

